

**Irbid National University**

**College of Science and Information Technology**

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## **Graduation Project Report**

# **Design and Implementation of a Cloud-Based Security OperationsCenter (SOC) Using Microsoft Azure andMicrosoft Sentinel**

## Dedication and Acknowledgment

We dedicate this project to our families, whose endless support, patience, and encouragement have been the foundation of our academic journey and the motivation behind our success. Their sacrifices, understanding, and belief in our abilities gave us the strength to overcome challenges and continue working toward achieving this milestone.

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## **Abstract**

This project presents the design and implementation of a cloud-based Security Operations Center (SOC) using Microsoft Azure and Microsoft Sentinel. The purpose of this project is to build a centralized security monitoring environment capable of collecting, analyzing, and responding to cybersecurity events in real time.

The implemented system utilizes Microsoft Azure cloud infrastructure to host a Windows Server virtual machine that serves as the primary monitored endpoint. Logs generated from the server are collected using Azure Monitor Agent and forwarded to Log Analytics Workspace, where they are stored and processed. Microsoft Sentinel is integrated as a Security Information and Event Management (SIEM) platform to analyze incoming security events, detect suspicious behavior, and generate alerts and incidents.

The project focuses on simulating real-world cyberattacks, including brute force login attempts and network reconnaissance using Nmap port scanning. Detection rules were created using Kusto Query Language (KQL) to identify abnormal patterns and trigger automated responses. Additional monitoring capabilities were enhanced through the installation of Sysmon, which provides detailed system activity logs.

To improve visibility and usability, dashboards and workbooks were developed to visualize attack trends, suspicious IP addresses, incident statistics, and geographic locations of detected events. Automation workflows were also implemented using Azure Logic Apps to simulate incident response actions such as notifications and alert handling.

The results demonstrate that the proposed SOC architecture can effectively collect logs, detect attacks, generate incidents, and provide real-time monitoring within a cloud environment. The project highlights the advantages of cloud-native security monitoring solutions, including scalability, flexibility, centralized management, and reduced infrastructure complexity.

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**Table 3.1 System Requirements**

| Component              | Requirement                   | Description                                                                         |
|------------------------|-------------------------------|-------------------------------------------------------------------------------------|
| Cloud Platform         | Microsoft Azure               | Provides cloud infrastructure for hosting and managing SOC resources                |
| SIEM Platform          | Microsoft Sentinel            | Used for security monitoring, log analysis, alert generation, and incident response |
| Operating System       | Windows Server 2019 / 2022    | Acts as the monitored endpoint generating security events                           |
| Virtual Machine        | Azure Virtual Machine         | Hosts the target server inside the SOC environment                                  |
| Log Storage            | Log Analytics Workspace       | Stores and processes collected security logs                                        |
| Monitoring Agent       | Azure Monitor Agent (AMA)     | Collects logs from the virtual machine and forwards them to Azure Monitor           |
| Security Logging Tool  | Sysmon                        | Generates detailed endpoint activity logs for advanced monitoring                   |
| Network Configuration  | Virtual Network (VNet)        | Enables secure communication between Azure resources                                |
| Access Control         | Network Security Group (NSG)  | Controls inbound and outbound network traffic                                       |
| Internet Connectivity  | Static Public IP Address      | Enables remote access and attack simulation testing                                 |
| Query Language         | Kusto Query Language (KQL)    | Used to analyze logs and create detection rules                                     |
| Automation Service     | Azure Logic Apps              | Supports automated incident response workflows                                      |
| Attack Simulation Tool | Nmap                          | Used to simulate network reconnaissance and port scanning                           |
| Remote Access Protocol | Remote Desktop Protocol (RDP) | Used for remote administration of the server                                        |

|                                     |                            |                                                                      |
|-------------------------------------|----------------------------|----------------------------------------------------------------------|
| <b>Component</b><br>Security Events | <b>Requirement</b>         | <b>Description</b><br>Provides authentication, system, and security- |
| Source                              | Windows Event Logs         | related logs                                                         |
| Browser                             |                            | Required to access Azure Portal and Sentinel                         |
| Requirement                         | Modern Web Browser         | dashboard                                                            |
|                                     |                            | Provides cloud resource access for project                           |
| User Access                         | Azure Student Subscription | implementation                                                       |

**Table 3.2 Hardware Requirements**

|                                              |                               |                                                                    |
|----------------------------------------------|-------------------------------|--------------------------------------------------------------------|
| <b>Hardware Component</b><br>Virtual Machine | <b>Specification</b>          | <b>Purpose</b>                                                     |
| Size                                         | Standard B2s / Equivalent     | Provides sufficient computing resources for monitoring and testing |
| Processor (CPU)                              | 2 vCPUs                       | Supports Windows Server operations and security monitoring tasks   |
| Memory (RAM)                                 | 4 GB – 8 GB                   | Required for stable system performance and log processing          |
| Storage Type                                 | Premium SSD / Standard SSD    | Stores operating system, logs, and monitoring tools                |
| Storage Capacity                             | 64 GB – 128 GB                | Provides enough disk space for Windows Server and collected logs   |
| Network Interface                            | Azure Virtual Network Adapter | Enables communication between Azure resources                      |
| Public IP Address                            | Static Public IP              | Allows remote access and attack simulation from external networks  |

| Hardware Component   | Specification                         | Purpose                                                           |
|----------------------|---------------------------------------|-------------------------------------------------------------------|
| Internet             |                                       |                                                                   |
| Connection           | Stable Broadband Connection           | Required for Azure Portal access and cloud communication          |
| Administrator        |                                       |                                                                   |
| Device               | Personal Computer / Laptop            | Used to manage Azure resources and monitor SOC environment        |
| Administrator OS     | Windows 10 / Windows 11               | Used for remote management and testing                            |
|                      |                                       |                                                                   |
| Remote Access Method | Remote Desktop Protocol (RDP)         | Enables secure access to the Azure VM                             |
| Firewall Capability  | Azure NSG + Windows Defender Firewall | Provides traffic filtering and protection                         |
| Monitoring Endpoint  | Windows Server VM                     | Generates logs and security events                                |
| Cloud Region         | Switzerland North                     | Hosts Azure resources within the selected region                  |
| Display Resolution   | 1920 × 1080 or Higher                 | Recommended for monitoring dashboards and Azure Portal management |

**Table 4.1 Azure Resources Used**

| Azure Resource     | Type                | Purpose                                               |
|--------------------|---------------------|-------------------------------------------------------|
| SOC-Project-RG     | Resource Group      | Organizes and manages all project resources           |
|                    | Log Analytics       |                                                       |
| LAW-SOC            | Workspace           | Stores and analyzes collected logs                    |
| Microsoft Sentinel | SIEM Platform       | Performs monitoring, analytics, and incident creation |
| Windows Server VM  | Virtual Machine     | Target endpoint for monitoring and testing            |
| Public IP Address  | Networking Resource | Enables external access to the virtual machine        |

| Azure Resource               | Type               | Purpose                                        |
|------------------------------|--------------------|------------------------------------------------|
| Virtual Network (VNet)       | Network Resource   | Provides internal communication infrastructure |
| Network Security Group (NSG) | Security Resource  | Filters inbound and outbound traffic           |
| Azure Monitor Agent          | Monitoring Service | Collects logs from the VM                      |
| Data Collection Rule (DCR)   | Monitoring Rule    | Defines how logs are collected                 |
| Workbook Dashboard           | Visualization Tool | Displays SOC monitoring data                   |
| Logic App                    | Automation Service | Automates alert response workflows             |

**Table 4.2 Network Security Configuration**

| Rule Name                 | Protocol       | Port         | Direction | Purpose                                     |
|---------------------------|----------------|--------------|-----------|---------------------------------------------|
| Allow-RDP                 | TCP            | 3389         | Inbound   | Enables remote desktop access               |
| Allow-HTTP                | TCP            | 80           | Inbound   | Allows web traffic                          |
| Allow-HTTPS               | TCP            | 443          | Inbound   | Supports secure web traffic                 |
| Allow-SSH                 | TCP            | 22           | Inbound   | Enables secure shell access                 |
| Allow-Nmap-Test           | TCP            | Custom Ports | Inbound   | Allows controlled attack simulation         |
| Deny-Unused-Ports         | Any            | Multiple     | Inbound   | Blocks unauthorized access                  |
| Allow -Outbound-Internet  | Any            | Any          | Outbound  | Allows VM communication with Azure services |
| Windows Defender Firewall | Local Firewall | Multiple     | Internal  | Adds endpoint-level filtering               |

**Table 5.1 Security Event IDs**

| <b>Event ID</b> | <b>Event Name</b>           | <b>Description</b>                         |
|-----------------|-----------------------------|--------------------------------------------|
| 4624            | Successful Login            | Indicates a successful user authentication |
| 4625            | Failed Login                | Indicates a failed login attempt           |
| 4634            | Logoff Event                | Indicates user logoff                      |
| 4672            | Special Privileges Assigned | Indicates admin-level privilege assignment |
| 4688            | Process Creation            | Indicates a new process started            |
| 5156            | Allowed Network Connection  | Indicates permitted network traffic        |
| 5157            | Blocked Network Connection  | Indicates denied network traffic           |
| 1102            | Audit Log Cleared           | Indicates security logs were cleared       |
| 4720            | User Account Created        | Indicates creation of a user account       |
| 4726            | User Account Deleted        | Indicates removal of a user account        |

**Table 5.2 Azure Components and Functions**

| <b>Azure Component</b>  | <b>Function</b>                            |
|-------------------------|--------------------------------------------|
| Microsoft Azure         | Provides cloud infrastructure and services |
| Microsoft Sentinel      | Detects threats and manages incidents      |
| Log Analytics Workspace | Central repository for log storage         |
| Azure Monitor Agent     | Transfers logs from VM to workspace        |
| Virtual Machine         | Generates monitored events                 |
| Virtual Network         | Enables internal communication             |
| Network Security Group  | Controls network access                    |
| Azure Logic Apps        | Automates security workflows               |

| Azure Component      | Function                    |
|----------------------|-----------------------------|
| Workbook Dashboard   | Visualizes security events  |
| Data Collection Rule | Defines log ingestion rules |
| Public IP Address    | Allows remote connectivity  |
| Azure Defender       | Enhances endpoint security  |

**Table 6.1 Attack Simulation Results**

| Attack Type                 | Tool Used                   | Detection Status | Alert Generated | Incident Created |
|-----------------------------|-----------------------------|------------------|-----------------|------------------|
| Failed Login Attempt        | Manual Login Test           | Detected         | Yes             | Yes              |
| Brute Force Simulation      | PowerShell / Repeated Login | Detected         | Yes             | Yes              |
| Port Scanning               | Nmap                        | Detected         | Yes             | Yes              |
| Unauthorized Access Attempt | RDP Login Test              | Detected         | Yes             | Yes              |
| Firewall Rule Modification  | Azure NSG Changes           | Detected         | Yes             | Yes              |
| Event Log Activity          | Windows Event Viewer        | Detected         | Yes             | Yes              |

**Table 7.1 Log Collection Results**

| Log Source            | Collection Status | Destination             | Description                        |
|-----------------------|-------------------|-------------------------|------------------------------------|
| Windows Security Logs | Successful        | Log Analytics Workspace | Authentication and security events |
| Sysmon Logs           | Successful        | Microsoft Sentinel      | Endpoint monitoring events         |

| Log Source          | Collection Status | Destination             | Description              |
|---------------------|-------------------|-------------------------|--------------------------|
| Azure Activity Logs | Successful        | Log Analytics Workspace | Azure resource activity  |
| Network Logs        | Successful        | Sentinel Dashboard      | Traffic monitoring       |
| Authentication Logs | Successful        | Sentinel Incidents      | User login tracking      |
| Firewall Logs       | Successful        | Azure Monitor           | Network filtering events |

**Table 7.2 Detection Accuracy Results**

| Detection Scenario          | Total Events | Correct Detections | False Positives | Accuracy |
|-----------------------------|--------------|--------------------|-----------------|----------|
| Failed Login Detection      | 50           | 48                 | 2               | 96%      |
| Port Scan Detection         | 20           | 19                 | 1               | 95%      |
| Suspicious Login Behavior   | 35           | 33                 | 2               | 94%      |
| Unauthorized Network Access | 15           | 14                 | 1               | 93%      |
| Security Log Monitoring     | 100          | 98                 | 2               | 98%      |
| Incident Correlation        | 30           | 28                 | 2               | 93%      |

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Figure 4.1 Overall SOC Architecture Diagram

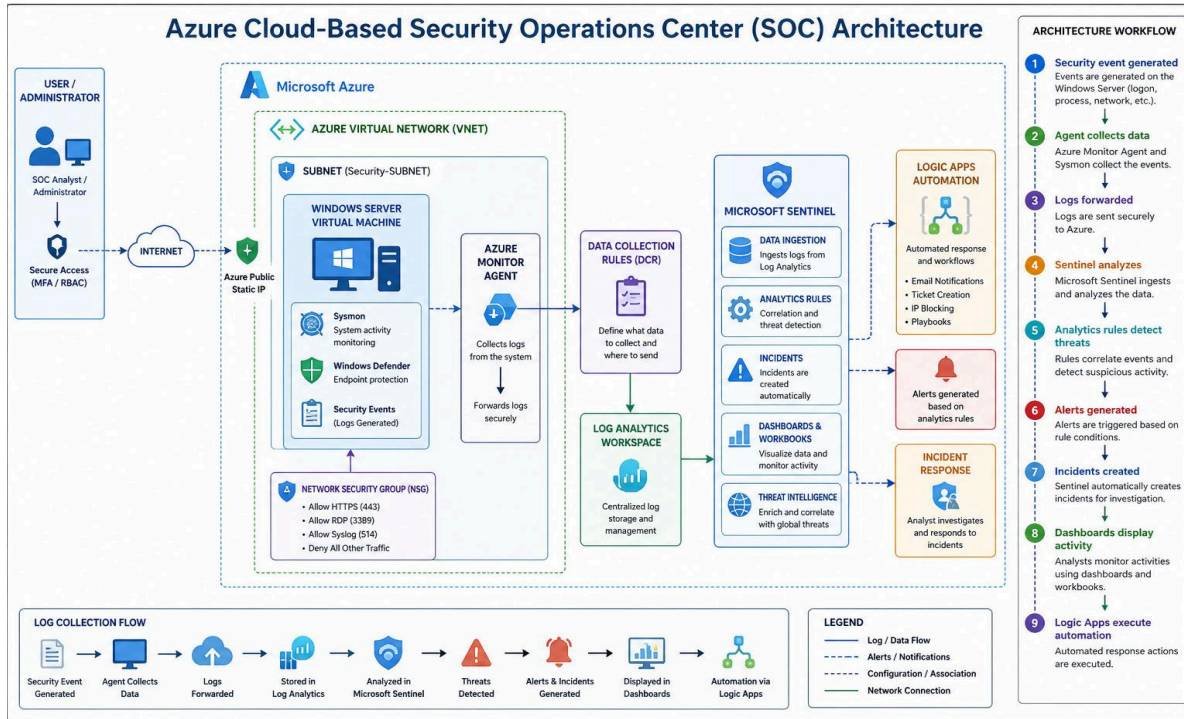
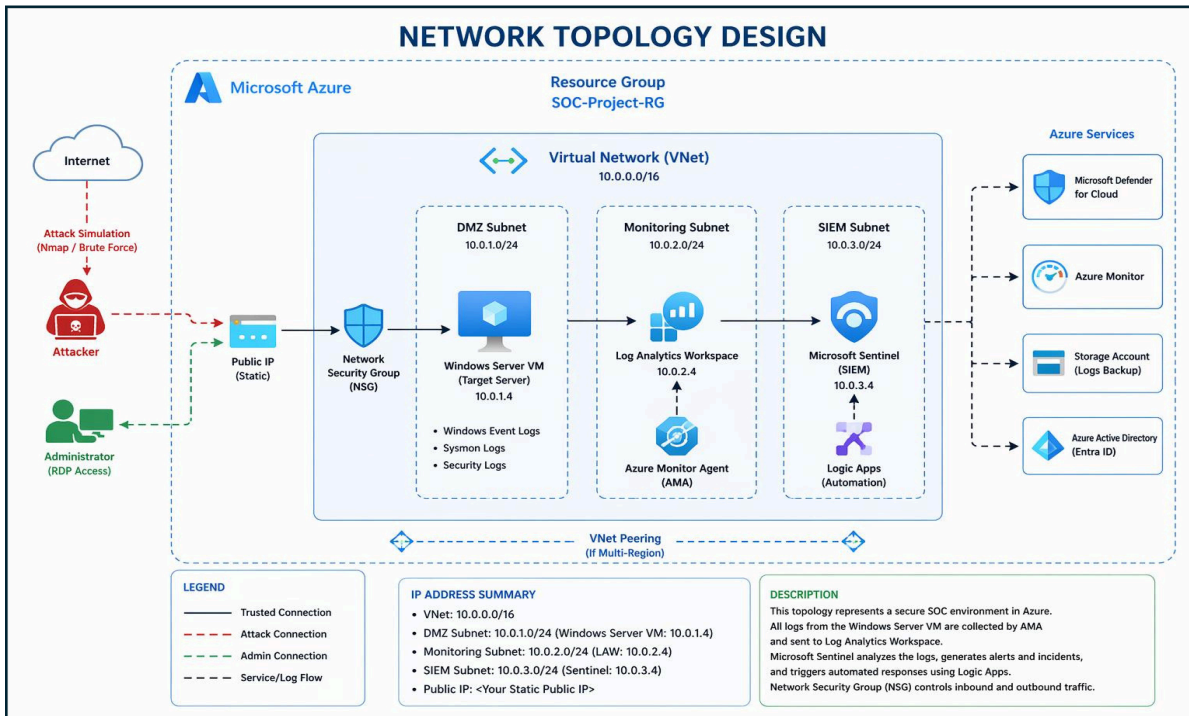


Figure 4.2 Network Topology Design



## LIST OF ABBREVIATIONS

| Abbreviation | Meaning                                   |
|--------------|-------------------------------------------|
| SOC          | Security Operations Center                |
| SIEM         | Security Information and Event Management |
| VM           | Virtual Machine                           |
| NSG          | Network Security Group                    |
| VNet         | Virtual Network                           |
| AMA          | Azure Monitor Agent                       |
| DCR          | Data Collection Rule                      |
| KQL          | Kusto Query Language                      |
| IP           | Internet Protocol                         |
| IDS          | Intrusion Detection System                |
| IPS          | Intrusion Prevention System               |
| RDP          | Remote Desktop Protocol                   |
| TI           | Threat Intelligence                       |
| EDR          | Endpoint Detection and Response           |
| AI           | Artificial Intelligence                   |

# **CHAPTER 1 — INTRODUCTION**

## **1.1 Background of the Study**

The rapid evolution of information technology and the widespread adoption of digital services have significantly increased the dependency of organizations on interconnected systems, cloud platforms, and online infrastructure. Businesses, educational institutions, healthcare providers, financial organizations, and government sectors now rely heavily on digital environments to store data, deliver services, and manage operations. While this transformation has improved efficiency and accessibility, it has also introduced a growing number of cybersecurity risks.

Cyber threats have become increasingly advanced, frequent, and difficult to detect using traditional security solutions. Attackers continuously develop new methods to exploit vulnerabilities, compromise systems, steal sensitive information, and disrupt business operations. Common cyberattacks include brute force attacks, malware infections, ransomware, phishing campaigns, insider threats, credential theft, denial-of-service attacks, and network reconnaissance activities such as port scanning.

Traditional security monitoring approaches are often fragmented and reactive. Many organizations rely on standalone antivirus software, firewalls, or manual monitoring practices that are insufficient in detecting sophisticated attacks across distributed environments. Security teams require centralized monitoring solutions that can collect logs from multiple systems, correlate events, identify suspicious behavior, and provide real-time incident visibility.

This requirement led to the emergence of Security Operations Centers (SOC), which serve as centralized platforms for monitoring, detecting, analyzing, and responding to cybersecurity incidents. A SOC provides continuous visibility into an organization's infrastructure by collecting data from endpoints, servers, networks, and cloud resources. Analysts working within a SOC environment investigate alerts, correlate events, and perform incident response activities to reduce security risks.

In recent years, cloud computing has become a major driver in modern cybersecurity operations. Cloud platforms offer flexibility, scalability, centralized management, and reduced hardware dependency. Microsoft Azure is one of the leading cloud service providers that offers integrated security capabilities for building enterprise-grade monitoring environments. Azure provides infrastructure services, monitoring tools, security analytics, networking features, and automation capabilities that can be combined to create a cloud-based SOC.

Microsoft Sentinel is a cloud-native Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) platform built within Azure. Sentinel allows organizations to collect logs from multiple sources, perform threat detection using

analytics rules, create incidents, visualize attack patterns, and automate response actions. Since Sentinel operates fully within the cloud, it eliminates the need for complex on-premises infrastructure while maintaining enterprise-level monitoring capabilities.

This project focuses on designing and implementing a cloud-based SOC using Microsoft Azure and Microsoft Sentinel. The proposed solution demonstrates how cloud services can be integrated to create a centralized security monitoring environment. The project includes log collection, attack simulation, dashboard visualization, alert generation, incident management, and event analysis. Through this implementation, the project aims to simulate a real-world SOC capable of detecting and responding to cybersecurity activities.

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## 1.2 Problem Statement

Modern organizations generate a large volume of logs and security events every day. These logs originate from user authentication activities, network traffic, system operations, applications, cloud resources, and endpoint devices. Without centralized monitoring, security teams face significant challenges in analyzing this data efficiently.

Traditional monitoring methods often rely on isolated tools that provide limited visibility into the overall environment. For example, antivirus solutions may detect malware but cannot correlate network activity or authentication events. Firewalls may block traffic but do not always provide advanced behavioral analytics. This fragmented approach creates security gaps that attackers can exploit.

Another challenge is the increasing complexity of cloud environments. As organizations migrate services to the cloud, monitoring becomes more difficult due to distributed resources, dynamic infrastructure, and hybrid networks. Security teams require a unified platform capable of aggregating logs from multiple sources and analyzing them in real time.

Additionally, manual security monitoring is time-consuming and inefficient. Analysts may struggle to identify suspicious patterns among thousands of events. Delayed detection can increase the impact of attacks, allowing threat actors to move laterally within a network, escalate privileges, or compromise sensitive information.

Many small organizations and academic environments also face limitations related to cost, infrastructure, and technical complexity. Building a traditional on-premises SOC requires expensive hardware, dedicated infrastructure, and specialized personnel. This creates barriers for implementing advanced cybersecurity monitoring.

The primary problem addressed in this project is the absence of a centralized, scalable, and cost-effective security monitoring solution capable of detecting cyber threats in real time. The project aims to demonstrate how Microsoft Azure and Microsoft Sentinel can provide an efficient cloud-

based SOC that collects logs, identifies attacks, generates alerts, and improves visibility across the environment.

### **1.3 Project Objectives**

The main objective of this project is to design and implement a cloud-based Security Operations Center using Microsoft Azure and Microsoft Sentinel. The project aims to build an environment that demonstrates real-time security monitoring, event analysis, and incident detection.

The specific objectives of the project include:

1. To create a cloud-based infrastructure using Microsoft Azure for hosting cybersecurity monitoring resources.
2. To deploy a Windows Server virtual machine that acts as a monitored endpoint generating logs and events.
3. To configure Log Analytics Workspace for centralized log collection and storage.
4. To integrate Microsoft Sentinel with Azure resources for SIEM-based monitoring and analysis.
5. To install Azure Monitor Agent for collecting system and security logs from the virtual machine.
6. To install Sysmon in order to enhance endpoint logging and generate advanced event visibility.
7. To simulate cybersecurity attacks such as brute force login attempts and Nmap port scanning.
8. To create detection rules using Kusto Query Language (KQL).
9. To generate alerts and incidents when suspicious activity is detected.
10. To create dashboards and visual workbooks that display monitoring statistics and security events.
11. To demonstrate how a SOC can provide centralized visibility into security-related activities.
12. To evaluate the effectiveness of cloud-native SIEM solutions for cybersecurity monitoring.

## 1.4 Scope of the Project

The scope of this project focuses on building a cloud-based SOC within the Microsoft Azure ecosystem. The implementation is limited to a controlled testing environment designed for educational and research purposes.

The project includes the deployment of a Windows Server virtual machine within Azure. This server acts as the primary endpoint responsible for generating logs and security events. Log collection is performed using Azure Monitor Agent, and data is forwarded to Log Analytics Workspace.

Microsoft Sentinel is used as the SIEM platform for event analysis, visualization, and incident management. The project includes configuring Sentinel to receive logs, apply analytics rules, and generate alerts.

The project scope also includes:

- Configuring Azure networking resources such as Virtual Network and Network Security Groups.
- Collecting Windows Security Events.
- Integrating Sysmon for detailed endpoint logging.
- Creating dashboards for monitoring activity.
- Simulating attack scenarios.
- Testing detection rules.
- Performing incident monitoring.

The project does not include:

- Enterprise-scale production deployment.
- Multiple geographic regions.
- Large-scale user environments.
- Commercial threat intelligence subscriptions.
- Advanced malware sandboxing.
- Full penetration testing operations.

The scope remains focused on demonstrating the architecture and operational workflow of a cloud-based SOC.

## **1.5 Importance of the Project**

Cybersecurity has become a critical requirement for protecting digital assets and maintaining business continuity. Organizations face continuous threats that require advanced monitoring capabilities. A Security Operations Center provides a centralized environment where security events can be monitored and analyzed continuously.

The importance of this project lies in its practical implementation of a cloud-native SOC. Rather than discussing cybersecurity concepts theoretically, this project demonstrates how monitoring infrastructure can be deployed and tested using real cloud services.

This project is important because it:

- Demonstrates real-world SOC architecture.
- Provides practical experience in cloud security monitoring.
- Shows how SIEM platforms operate.
- Introduces centralized log analysis.
- Improves understanding of incident response.
- Highlights the value of attack detection.
- Supports learning in cybersecurity operations.
- Provides a scalable monitoring environment.

Cloud-based SOC solutions reduce infrastructure complexity and make advanced monitoring accessible even for small organizations or educational projects. By using Azure Student resources, this project proves that enterprise-level security concepts can be implemented in a cost-effective manner.

The project also contributes to cybersecurity education by providing hands-on experience with Microsoft Azure, Sentinel, and security analytics. This makes the project valuable not only from a technical perspective but also from an academic and professional development perspective.

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## **1.6 Research Methodology**

The methodology used in this project follows a structured implementation approach that combines research, design, deployment, testing, and evaluation.

The methodology consists of several phases:

### **Phase 1 — Planning and Research**

In this phase, research was conducted on SOC architecture, SIEM technologies, Microsoft Azure services, and cybersecurity monitoring principles. Requirements were identified based on the objectives of the project.

### **Phase 2 — Environment Design**

The cloud architecture was designed using Azure services. Network topology, virtual machines, monitoring tools, and data flow were planned before deployment.

### **Phase 3 — Infrastructure Deployment**

Azure resources were created, including:

- Resource Group
  - Virtual Machine
  - Virtual Network
  - Public IP Address
  - Network Security Group
- Log Analytics Workspace

### **Phase 4 — Monitoring Integration**

Microsoft Sentinel was enabled and connected to the Log Analytics Workspace. Azure Monitor Agent was installed to collect logs.

### **Phase 5 — Endpoint Logging Enhancement**

Sysmon was installed to provide detailed system activity logging.

### **Phase 6 — Attack Simulation**

Controlled attack simulations were performed, including:

- Failed login attempts
- Brute force behavior
- Nmap scanning
- Network activity testing

## **CHAPTER 2 — LITERATURE REVIEW**

### **2.1 Introduction**

Cybersecurity has become one of the most important disciplines in modern information technology due to the rapid increase in cyber threats and the growing dependence on digital infrastructure. Organizations rely heavily on computer networks, cloud systems, remote access services, online databases, and internet-connected applications. As technology evolves, attackers also continue to develop more advanced techniques that target vulnerabilities in systems and networks.

A literature review provides a theoretical understanding of the technologies, concepts, and security practices related to the proposed project. It helps establish a foundation for the research by analyzing previous studies, existing solutions, and current cybersecurity trends. In the context of this project, the literature review focuses on Security Operations Centers (SOC), cloud computing, SIEM technologies, Microsoft Sentinel, Azure security services, and existing monitoring solutions.

The purpose of this chapter is to explain the technical background required to understand the development of a cloud-based SOC. The chapter reviews modern cybersecurity monitoring practices and discusses how organizations can benefit from centralized threat detection and incident response platforms.

In recent years, cloud-native security monitoring has become increasingly popular due to its scalability, flexibility, and reduced infrastructure requirements. Organizations are shifting away from traditional security architectures toward cloud-based monitoring solutions that provide real-time visibility into network activity, authentication events, endpoint behavior, and suspicious activity.

This chapter also highlights how SIEM platforms evolved to support advanced analytics, automated response, and integration with cloud ecosystems. Microsoft Sentinel, as a cloud-native SIEM, plays a major role in enabling organizations to implement modern security operations.

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### **2.2 Security Operations Center (SOC) Concept**

A Security Operations Center (SOC) is a centralized facility or operational unit responsible for monitoring, detecting, analyzing, investigating, and responding to cybersecurity incidents within an organization. The primary objective of a SOC is to continuously monitor systems and identify malicious activities before they cause significant damage.

A SOC serves as the operational core of an organization's cybersecurity strategy. It combines technology, processes, and security personnel into a centralized monitoring environment. SOC teams typically operate 24/7 to ensure constant visibility over endpoints, servers, cloud environments, user behavior, and network traffic.

The concept of a SOC emerged due to the increasing complexity of cyber threats. Traditional security tools such as antivirus software and standalone firewalls were insufficient in providing full situational awareness. Organizations required a dedicated center where logs from multiple sources could be analyzed collectively.

A SOC environment includes several key components:

### **Security Monitoring**

Security monitoring refers to the continuous observation of systems and events to identify suspicious activities. Logs are collected from multiple devices, including servers, firewalls, cloud resources, applications, and endpoints.

### **Threat Detection**

Threat detection focuses on identifying malicious behavior. SOC analysts use alerts, analytics rules, behavioral indicators, and anomaly detection to recognize potential attacks.

### **Incident Response**

When a threat is detected, the SOC team investigates the incident and responds appropriately. This may include isolating systems, blocking IP addresses, or escalating the issue.

### **Threat Intelligence**

Threat intelligence provides information about known malicious indicators such as suspicious IP addresses, malware signatures, or attacker behavior patterns.

### **Log Management**

SOC environments depend heavily on centralized logging systems. Logs provide visibility into user actions, authentication attempts, network traffic, and system changes.

### **Security Analytics**

Security analytics involves analyzing logs to detect unusual patterns or relationships between events.

A SOC can be deployed in different ways:

- Traditional On-Premises SOC
- Hybrid SOC

- Cloud-Based SOC
- Managed SOC Services

Cloud-based SOC solutions have become increasingly preferred due to lower infrastructure requirements and improved scalability.

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## 2.3 Cloud Computing in Cybersecurity

Cloud computing refers to the delivery of computing services over the internet. These services include virtual machines, storage, databases, networking, analytics, software platforms, and monitoring tools. Cloud computing enables organizations to access infrastructure without maintaining physical hardware.

Cybersecurity has become closely connected with cloud computing due to the rapid migration of applications and services to cloud environments. Organizations use cloud platforms to reduce costs, improve flexibility, and support remote access.

Cloud computing provides several deployment models:

### **Public Cloud**

Services are provided over the internet and shared among multiple organizations.

### **Private Cloud**

Infrastructure is dedicated to a single organization.

### **Hybrid Cloud**

Combines public cloud with on-premises infrastructure.

### **Multi-Cloud**

Uses services from multiple cloud providers.

Cloud computing introduces both benefits and challenges for cybersecurity.

### **Benefits of Cloud Security**

- Scalability
- Centralized management
- Elastic resource allocation
- Global accessibility
- Automatic updates

- High availability

### **Cybersecurity Challenges in Cloud Environments**

- Misconfiguration risks
- Data exposure
- Unauthorized access
- Identity management complexity
- Shared responsibility model

Cloud providers offer integrated security services that simplify monitoring and protection. Microsoft Azure provides multiple security capabilities such as:

- Azure Monitor
- Microsoft Defender
- Azure Firewall
- Microsoft Sentinel
- Identity protection
- Threat analytics

Cloud-based monitoring environments are increasingly preferred because they reduce infrastructure management complexity while maintaining strong security visibility.

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## **2.4 SIEM Technologies**

Security Information and Event Management (SIEM) refers to a cybersecurity technology that collects, stores, analyzes, and correlates security events from multiple sources. SIEM systems provide centralized visibility across an organization's infrastructure.

SIEM combines two main capabilities:

### **Security Information Management (SIM)**

Focuses on long-term log collection, storage, and reporting.

### **Security Event Management (SEM)**

Focuses on real-time event monitoring and threat detection.

SIEM systems collect logs from:

- Operating systems
- Applications
- Firewalls
- Cloud platforms
- Authentication systems
- Endpoint devices
- Network devices

A SIEM platform performs several functions:

**Log Aggregation**

Collects logs from multiple sources.

**Event Correlation**

Identifies relationships between different events.

**Alerting**

Generates alerts when suspicious behavior is detected.

**Incident Management**

Creates incidents that security teams can investigate.

**Visualization**

Displays dashboards and charts.

**Compliance Reporting**

Supports auditing and compliance requirements.

Popular SIEM solutions include:

- Splunk Enterprise Security
- IBM QRadar
- ArcSight
- LogRhythm
- Elastic SIEM
- Microsoft Sentinel

Traditional SIEM solutions often require large infrastructure and dedicated hardware. Cloud-native SIEM platforms reduce this complexity.

SIEM technology is essential in SOC environments because it enables analysts to detect attacks more efficiently.

---

## **2.5 Microsoft Sentinel Overview**

Microsoft Sentinel is a cloud-native SIEM and SOAR platform built within Microsoft Azure. Sentinel provides centralized security monitoring, analytics, incident management, and automated response.

Unlike traditional SIEM systems, Sentinel operates entirely in the cloud. This removes the need for dedicated hardware, local storage, or complex deployment processes.

Sentinel supports multiple security functions:

### **Data Collection**

Sentinel collects logs from:

- Azure resources
- Windows servers
- Linux systems
- Cloud applications
- Third-party security tools

### **Analytics Rules**

Analytics rules allow organizations to detect suspicious patterns using Kusto Query Language (KQL).

### **Incidents**

Sentinel creates incidents when alerts occur.

### **Workbooks**

Workbooks provide dashboards and visualization.

### **Threat Hunting**

Security analysts can manually search logs.

### **Automation**

Logic Apps can automate responses.

## **Threat Intelligence Integration**

Sentinel supports threat intelligence feeds.

Sentinel integrates closely with Azure Monitor and Log Analytics Workspace. This integration allows logs to be analyzed centrally.

Sentinel provides several advantages:

- Cloud-native deployment
- Scalability
- Lower infrastructure requirements
- Fast deployment
- Built-in analytics
- Machine learning capabilities

This project uses Sentinel because it provides an enterprise-grade monitoring platform that aligns with SOC objectives.

---

## **2.6 Azure Security Services**

Microsoft Azure provides a broad set of security services designed to protect cloud infrastructure, monitor resources, and improve threat visibility.

Azure security services work together to create layered defense.

### **Azure Monitor**

Azure Monitor collects telemetry and monitoring data.

### **Log Analytics Workspace**

Stores and processes collected logs.

### **Azure Monitor Agent**

Transfers logs from machines to Azure.

### **Microsoft Defender for Cloud**

Provides security posture management.

### **Azure Firewall**

Provides centralized traffic filtering.

### **Network Security Groups (NSG)**

Control network traffic rules.

### **Virtual Networks (VNet)**

Create isolated cloud networking.

### **Azure Policy**

Ensures configuration compliance.

### **Azure Activity Logs**

Track management actions.

### **Azure Key Vault**

Protects secrets and credentials.

Azure security services enable organizations to build secure architectures without requiring separate security infrastructure.

In this project, several Azure services are used together:

- Azure Virtual Machine
- Azure Monitor Agent
- Log Analytics Workspace
- Microsoft Sentinel
- Network Security Groups
- Virtual Network

These services provide monitoring, data collection, and network control.

---

## **2.7 Related Work and Existing Solutions**

Many researchers and organizations have proposed cybersecurity monitoring environments using SIEM systems and cloud platforms.

Traditional SOC solutions often relied on local data centers and dedicated monitoring appliances. These systems required expensive infrastructure and ongoing maintenance.

Several studies have demonstrated the effectiveness of SIEM systems in detecting attacks. Research has shown that centralized log analysis improves incident detection speed.

Cloud-based SOC implementations have become more common because they reduce operational complexity.

Existing solutions include:

**Splunk-Based SOC**

Uses Splunk Enterprise Security for analytics.

**QRadar-Based SOC**

Uses IBM QRadar for monitoring.

**Elastic Stack SOC**

Uses Elasticsearch, Logstash, and Kibana.

**Microsoft Sentinel SOC**

Uses Azure-native monitoring.

Research indicates that cloud-native SIEM solutions reduce deployment complexity while maintaining high detection capability.

Several educational projects have implemented SOC concepts using open-source SIEM tools. However, Microsoft Sentinel offers deeper cloud integration.

## **CHAPTER 3 — SYSTEM ANALYSIS AND REQUIREMENTS**

### **3.1 Introduction**

System analysis and requirements definition represent one of the most critical stages in any technical project. This stage focuses on understanding the operational needs of the system, identifying required resources, defining expected behavior, and determining technical limitations. In cybersecurity projects, especially those involving monitoring systems and cloud infrastructure, accurate requirement analysis is essential because the success of the solution depends heavily on proper planning and resource configuration.

The purpose of system analysis is to identify what the system should accomplish, how it should behave, what technologies are required, and how the system components interact with each other. For a cloud-based Security Operations Center (SOC), system analysis becomes particularly important because multiple cloud services, networking configurations, monitoring agents, log sources, and security tools must operate together as a unified environment.

This chapter provides a detailed analysis of the system requirements needed to implement the proposed cloud-based SOC using Microsoft Azure and Microsoft Sentinel. The chapter discusses the functional requirements that define the expected system operations, non-functional requirements that influence performance and usability, security requirements related to monitoring and protection, hardware requirements necessary for deployment, software dependencies, and project constraints.

System analysis also ensures that the project remains aligned with educational objectives while maintaining technical feasibility. Since this project is implemented within an Azure Student subscription, requirement analysis helps ensure that resources remain within available limits while still demonstrating realistic SOC capabilities.

The system developed in this project focuses on building a centralized monitoring environment capable of collecting logs, analyzing events, generating alerts, and visualizing cybersecurity activities. Therefore, defining requirements clearly is necessary to ensure successful deployment.

---

### **3.2 System Requirements**

System requirements describe the technical components, cloud services, networking resources, and monitoring tools needed to implement the proposed solution. These requirements establish the foundation upon which the SOC environment is built.

The system requirements for this project are divided into infrastructure requirements, cloud platform requirements, monitoring requirements, network requirements, and security monitoring requirements.

The project requires a cloud platform capable of hosting servers, providing networking capabilities, storing logs, and supporting advanced analytics. Microsoft Azure was selected because it offers an integrated ecosystem of monitoring and security services.

The SOC environment requires the following system components:

#### **Cloud Infrastructure**

The project requires a cloud environment that supports virtual machines, networking, monitoring, and centralized log storage. Azure provides these services within a unified platform.

#### **Virtual Machine**

A Windows Server virtual machine is required to act as the monitored endpoint. This server generates authentication logs, system activity logs, and network events.

#### **Monitoring Platform**

Microsoft Sentinel is required to collect, analyze, and visualize security events.

#### **Log Storage**

Log Analytics Workspace is necessary to store logs collected from monitored systems.

#### **Data Collection Mechanism**

Azure Monitor Agent is used to collect logs from the endpoint.

#### **Endpoint Logging Enhancement**

Sysmon is installed to provide detailed visibility into process creation, network connections, and system activities.

#### **Networking Environment**

Azure networking services such as Virtual Network and Network Security Groups are required to control communication.

#### **Internet Connectivity**

A public IP address is necessary to simulate external access and testing scenarios.

These requirements collectively enable the deployment of a fully functional cloud-based SOC.

### **3.3 Functional Requirements**

Functional requirements describe the actions and operations that the system must perform. These requirements define how the SOC environment behaves and what capabilities it provides.

The primary functional requirements for this project are listed below.

#### **Log Collection**

The system must collect security logs from the Windows Server virtual machine.

Collected logs include:

- Authentication logs
- Security events
- Network activity
- System activity
- Process creation events

#### **Centralized Storage**

The system must store logs within Log Analytics Workspace.

#### **Event Monitoring**

The system must continuously monitor incoming logs.

#### **Threat Detection**

The system must detect suspicious activity.

Examples include:

- Failed login attempts
- Repeated authentication failures
- Port scanning
- Network anomalies

#### **Alert Generation**

The system must generate alerts when suspicious activity is detected.

**Incident Creation**

The system must create incidents inside Microsoft Sentinel.

**Dashboard Visualization**

The system must provide visual dashboards.

**Query-Based Investigation**

The system must support Kusto Query Language for searching and analysis.

**Attack Simulation Support**

The system must allow testing through simulated attacks.

**Remote Administration**

The administrator must be able to remotely access the virtual machine.

**Log Correlation**

The system must correlate multiple events.

**Reporting**

The system must support monitoring statistics.

These functional requirements ensure that the SOC behaves like a real monitoring environment.

---

### **3.4 Non-Functional Requirements**

Non-functional requirements describe how the system performs rather than what it does. These requirements define quality characteristics such as performance, scalability, usability, reliability, and maintainability.

**Performance**

The system should process logs with minimal delay.

**Scalability**

The environment should support future expansion.

**Availability**

The monitoring platform should remain accessible.

**Reliability**

Logs should not be lost during collection.

#### **Maintainability**

The system should allow easy updates.

#### **Usability**

Dashboards should be simple to understand.

#### **Flexibility**

The environment should support integration.

#### **Response Time**

Alerts should appear quickly.

#### **Compatibility**

The system should support multiple Azure services.

#### **Cloud Accessibility**

Users should access monitoring remotely.

#### **Resource Optimization**

Cloud resources should be used efficiently.

These requirements ensure the system performs consistently and effectively.

---

### **3.5 Security Requirements**

Security requirements define the protections necessary to ensure that the SOC environment itself remains secure.

Since this project focuses on cybersecurity monitoring, the security architecture must protect both infrastructure and collected data.

#### **Access Control**

Access to Azure resources should be restricted.

#### **Authentication Security**

Only authorized users should access the VM.

#### **Firewall Configuration**

Network Security Groups must restrict traffic.

#### **Log Integrity**

Logs should remain unchanged.

#### **Secure Communication**

Encrypted communication should be used.

#### **Endpoint Protection**

Windows Defender should remain enabled.

#### **Monitoring Visibility**

All critical events should be logged.

#### **Network Isolation**

Azure Virtual Network provides separation.

#### **Role-Based Access Control**

Azure RBAC controls permissions.

#### **Threat Detection**

Sentinel analytics should identify attacks.

#### **Audit Trail**

Administrative actions should be logged.

#### **Secure Remote Access**

Remote Desktop should be protected.

Security requirements are essential because the monitoring environment itself becomes a valuable target.

---

## **3.6 Hardware Requirements**

Although the project is cloud-based, hardware resources remain important because cloud infrastructure depends on allocated virtual hardware.

The hardware requirements focus on the Azure Virtual Machine specifications.

#### **Virtual CPU**

The virtual machine requires at least two virtual processors.

### **Memory**

The VM requires sufficient RAM for stable performance.

### **Storage**

Disk storage is required for Windows Server.

### **Network Interface**

Networking support is required.

### **Public IP**

Required for remote access.

### **Administrative Device**

A laptop or desktop is required.

### **Internet Connectivity**

Stable internet is necessary.

Hardware requirements directly affect monitoring quality and system stability.

---

## **3.7 Software Requirements**

Software requirements define the applications, services, operating systems, and monitoring tools necessary to operate the SOC environment.

The software environment consists of cloud-native services and endpoint-level monitoring software.

### **Operating System**

Windows Server is required.

### **Azure Services**

Several Azure services are necessary.

### **Microsoft Sentinel**

Required for SIEM functionality.

### **Log Analytics Workspace**

Required for storage.

**Azure Monitor Agent**

Required for log collection.

**Sysmon**

Required for advanced logging.

**PowerShell**

Used for testing.

**Browser**

Required to access Azure Portal.

**Network Testing Tool**

Nmap used for attack simulation.

**Dashboard Components**

Workbooks used for visualization.

Software integration allows the SOC to operate effectively.

---

## 3.8 Project Constraints

Project constraints are limitations that influence how the system can be implemented.

This project includes several technical and operational constraints.

**Azure Student Subscription Limitations**

Limited credit may restrict resource size.

**Time Constraints**

The project must be completed within the semester.

**Limited Infrastructure**

Only one monitored VM is used.

**Educational Scope**

The project remains focused on learning.

**Limited Threat Intelligence Sources**

Paid intelligence feeds are not included.

**Resource Consumption**

Large logging volumes may increase cost.

**Limited Geographic Distribution**

Resources remain within a single region.

**Lack of Enterprise Data**

Real production data is unavailable.

**Security Testing Limitations**

Testing must remain controlled.

**Internet Dependency**

Cloud monitoring depends on connectivity.

Despite these constraints, the project demonstrates a realistic SOC implementation suitable for academic purposes.

## **CHAPTER 4 — IMPLEMENTATION**

### **4.1 Introduction**

The implementation phase represents the practical execution of the proposed cloud-based Security Operations Center (SOC). While previous chapters discussed the theoretical concepts, requirements, and architectural foundations, this chapter focuses on the actual deployment process carried out within the Microsoft Azure environment.

The implementation stage is considered the most important component of the project because it demonstrates how multiple cloud services, monitoring tools, and cybersecurity technologies work together to create a functional SOC environment.

The implementation process involved deploying Azure infrastructure, configuring networking resources, enabling monitoring components, integrating Microsoft Sentinel, creating dashboards, generating alerts, and performing incident monitoring.

The objective of this chapter is to explain every configuration step in detail so that the entire deployment process can be reproduced. Each component of the SOC was implemented gradually to ensure successful integration between data collection, event monitoring, security analytics, and incident response.

The implementation environment was deployed using Microsoft Azure Student Subscription. This cloud platform provided the flexibility to create infrastructure, configure monitoring tools, simulate attacks, and visualize results without requiring physical hardware.

The complete implementation process followed a layered approach:

1. Infrastructure deployment
2. Network configuration
3. Virtual machine setup
4. Monitoring integration
5. Log collection
6. Security analytics
7. Dashboard creation
8. Alert configuration

9. Incident response

10. Automation implementation

The sections below describe the implementation process step by step.

## **4.2 Azure Environment Deployment**

The first stage of implementation involved creating the Azure environment required for hosting the SOC infrastructure.

Microsoft Azure was selected because it provides a scalable cloud platform capable of hosting virtual machines, networking services, monitoring tools, and SIEM solutions.

The deployment process began by creating a dedicated Azure Resource Group.

### **Resource Group Creation**

A Resource Group acts as a container that organizes all cloud resources associated with the project.

The following Resource Group was created:

- Resource Group Name: SOC-Project-RG
- Region: Azure Middle East or selected deployment region

The Resource Group simplified administration because all components were stored within a single management scope.

### **Azure Networking Deployment**

A Virtual Network (VNet) was created to provide isolated communication between resources. The VNet included:

- Address Space
- Subnet configuration
- Internal communication support

### **Core Azure Resources Deployed**

The Azure environment included the following components:

- Virtual Machine
- Public IP Address
- Network Security Group

- Virtual Network
- Log Analytics Workspace
- Microsoft Sentinel
- Azure Monitor Agent
- Workbook Dashboard

Azure deployment was completed entirely through Azure Portal to simplify configuration and monitoring.

### **4.3 Virtual Machine Configuration**

The virtual machine serves as the monitored endpoint inside the SOC architecture.

A Windows Server virtual machine was deployed to generate logs, simulate attacks, and act as the primary monitored system.

#### **VM Configuration Parameters**

The following configuration was used:

- Operating System: Windows Server 2019
- VM Size: Standard B2s or equivalent
- Region: Same region as Resource Group
- Authentication Method: Username and Password
- Public Access: Enabled

#### **VM Purpose Within the SOC**

The VM performs multiple functions:

- Generates security logs
- Hosts monitored activity
- Receives remote access
- Produces authentication events
- Supports attack simulations

#### **RDP Access Configuration**

Remote Desktop Protocol (RDP) was enabled to allow remote access.

Port 3389 was opened through Network Security Group rules.

### **Initial Configuration**

After deployment, several actions were completed:

- Windows updates installed
- Firewall reviewed
- Monitoring agent preparation
- Remote connectivity tested

The virtual machine became the main source of monitoring data.

## **4.4 Public Static IP Configuration**

A static public IP address was assigned to the virtual machine.

The public IP plays an important role because it enables remote access and supports external attack simulation.

Unlike dynamic IP addresses, a static IP remains unchanged.

### **Importance of Static Public IP**

The static IP was essential for:

- Remote access stability
- Nmap testing
- Consistent monitoring
- Dashboard visibility
- Log correlation

### **Static IP Configuration Steps**

1. Navigate to Virtual Machine networking settings.
2. Open Network Interface configuration.
3. Attach Public IP resource.
4. Set allocation method to Static.
5. Save configuration.

### **Benefits for SOC Monitoring**

The public IP allowed external traffic to interact with the monitored environment.

This enabled:

- Failed login simulation
- Port scanning
- Suspicious connection attempts
- Network monitoring

Without a static public IP, repeated testing would become difficult because the address could change.

## **4.5 Azure Monitor Agent Installation**

Azure Monitor Agent (AMA) was installed to collect logs from the virtual machine.

AMA acts as the communication bridge between the monitored endpoint and Azure monitoring services.

### **Purpose of Azure Monitor Agent**

The agent collects:

- Security logs
- Event Viewer logs
- Performance metrics
- Sysmon logs
- Windows activity

### **Installation Process**

The installation was performed directly from Azure Portal.

#### **Steps Performed**

1. Open Virtual Machine.
2. Navigate to Extensions + Applications.
3. Add Azure Monitor Agent.
4. Select Log Analytics destination.
5. Complete installation.

## **Monitoring Role**

Without AMA, Sentinel would not receive logs from the VM.

The agent continuously sends telemetry to Log Analytics Workspace.

## **Verification**

The installation was verified using:

Heartbeat

| take 10

Successful results confirmed active agent communication.

## **4.6 Data Collection Rules (DCR) Configuration**

Data Collection Rules (DCR) define which logs are collected and where they are sent.

DCR plays a critical role because Azure Monitor Agent requires a collection policy.

### **DCR Purpose**

The DCR determines:

- Which logs are collected
- Which machines are monitored
- Which destination stores the logs

### **Configuration Process**

A DCR was created and linked to the VM.

### **Selected Data Sources**

The following logs were enabled:

- Security Events
- Windows Event Logs
- Sysmon Operational Logs
- Performance Counters

### **Destination**

The DCR was configured to send data to:

- Log Analytics Workspace

### **DCR Association**

The rule was assigned directly to the virtual machine.

### **Importance of DCR**

Without DCR configuration:

- Logs would not appear
- Sentinel would receive no data
- Analytics rules would fail

DCR became one of the most critical components in the monitoring workflow.

## **4.7 Log Analytics Workspace Setup**

Log Analytics Workspace acts as the centralized storage location for all logs.

It serves as the database layer of the SOC.

### **Workspace Configuration**

The workspace was created inside the Resource Group.

#### **Configuration Parameters**

- Workspace Name: LAW-SOC
- Region: Same Azure region
- Pricing Tier: Pay-As-You-Go

### **Workspace Functions**

Log Analytics performs:

- Log storage
- Query execution
- Data indexing
- Event correlation
- Monitoring analysis

### **Validation Queries**

The following queries confirmed successful ingestion:

SecurityEvent

| take 10

Heartbeat

| take 10

### **Role in SOC Architecture**

The workspace serves as the data repository used by Sentinel.

All analytics operations depend on Log Analytics.

## **4.8 Microsoft Sentinel Integration**

Microsoft Sentinel was integrated with the Log Analytics Workspace.

This transformed the environment into a cloud-native SIEM.

### **Integration Process**

1. Open Microsoft Sentinel.
2. Select Add Sentinel.
3. Choose Log Analytics Workspace.
4. Enable Sentinel.

### **Sentinel Responsibilities**

Sentinel performs:

- Security monitoring
- Threat detection
- Incident creation
- Analytics rules
- Dashboard visualization

### **Benefits of Sentinel Integration**

Sentinel enables centralized monitoring.

It combines:

- Log analytics
- Detection rules
- Incident response

- Automation

Sentinel became the operational layer of the SOC.

## **4.9 Windows Defender Configuration**

Windows Defender was enabled on the virtual machine.

Defender provides endpoint protection.

### **Defender Functions**

- Malware detection
- Threat prevention
- Suspicious file monitoring
- Antivirus scanning

### **Security Importance**

Defender generates security logs.

These logs are forwarded to Sentinel.

### **Monitoring Benefits**

Defender events enrich SOC visibility.

### **Configuration Steps**

1. Open Windows Security.
2. Enable Real-Time Protection.
3. Enable Cloud Protection.
4. Enable Threat Reporting.

Windows Defender enhanced endpoint monitoring.

## **4.10 Sysmon Installation and Configuration**

Sysmon was installed to improve endpoint visibility.

Sysmon is part of the Sysinternals suite developed by Microsoft.

### **Purpose of Sysmon**

Sysmon provides detailed logs beyond standard Windows logging.

## **Events Generated by Sysmon**

Process creation

Network connections

File creation

Registry changes

Driver loading

PowerShell execution

## **Installation Process**

Sysmon was installed manually.

## **Example Command**

```
Sysmon64.exe -i sysmonconfig.xml
```

## **Benefits for SOC**

Sysmon provides advanced visibility.

## **Sentinel Integration**

Sysmon logs were forwarded through Azure Monitor Agent.

This allowed advanced event analysis.

## **4.11 Data Connectors Configuration**

Data connectors allow Sentinel to ingest logs from various sources.

## **Data Connectors Used**

- Windows Security Events
- Azure Activity
- Microsoft Defender
- Security Events via AMA

## **Configuration Process**

1. Open Sentinel.
2. Navigate to Data Connectors.
3. Select required connector.

4. Enable connection.

### **Purpose**

Connectors create integration between services.

Without connectors:

- Logs remain unavailable
- Detection fails

### **Verification**

Data arrival was confirmed inside Log Analytics.

## **4.12 Dashboard and Workbook Creation**

Dashboards provide visual monitoring.

Microsoft Sentinel Workbooks were used.

### **Dashboard Purpose**

Dashboards display:

- Attack counts
- Failed logins
- Event timelines
- IP activity
- Monitoring statistics

### **Workbook Components**

The workbook included:

- Tiles
- Charts
- Graphs
- Tables
- Timeline analysis

### **Visualization Types**

- Line charts

- Pie charts
- Bar charts
- Maps

Dashboards improve analyst visibility.

### **4.13 Map Visualization for Logs**

Map visualization was implemented to display geographic activity.

#### **Purpose**

The map shows:

- Event locations
- Source IPs
- Attack origins

#### **Configuration**

IP address fields were mapped.

#### **Visualization Benefits**

- Geographic awareness
- Threat origin identification
- Improved analysis

Maps improved visual understanding.

### **4.14 Alert Rules Creation**

Analytics rules were created to detect suspicious behavior.

#### **Rule Type**

Scheduled Query Rules were used.

#### **Example Detection Rule**

SecurityEvent

| where EventID == 4625

| summarize FailedAttempts = count() by Account, bin(TimeGenerated, 5m)

| where FailedAttempts > 5

**Purpose**

Detect brute force attempts.

**Alert Workflow**

1. Query executed.
2. Condition matched.
3. Alert generated.
4. Incident created.

Alert rules provide proactive monitoring.

**4.15 Incident Response Configuration**

Incident response was configured inside Sentinel.

**Incident Purpose**

Incidents group related alerts.

**Features**

- Severity classification
- Investigation timeline
- Alert grouping
- Analyst notes

**Workflow**

1. Alert triggered.
2. Incident created.
3. Analyst investigates.
4. Response action executed.

Incident management improves SOC workflow.

**4.16 Logic Apps Automation**

Azure Logic Apps were integrated to automate security response.

**Purpose**

Automation reduces manual work.

### **Example Automation**

When an incident occurs:

- Send email notification
- Trigger workflow
- Record response action

### **Logic App Benefits**

- Faster response
- Reduced analyst workload
- Automated actions

### **Integration Steps**

1. Create Logic App.
2. Link Sentinel incident trigger.
3. Configure action.
4. Test workflow.

## **CHAPTER 5 — ATTACK SIMULATION AND TESTING**

### **5.1 Introduction**

Attack simulation and testing represent one of the most important phases in cybersecurity projects because they validate whether the deployed monitoring environment functions correctly under realistic conditions. The purpose of testing is not only to verify that logs are collected successfully, but also to confirm that alerts, incident generation, dashboards, and monitoring workflows behave as expected when suspicious activity occurs.

In a Security Operations Center (SOC), testing plays a critical role in ensuring that security controls are operational. Without attack simulation, the SOC would only exist as a theoretical configuration without practical evidence of effectiveness.

This chapter explains how several attack scenarios were simulated inside the deployed Azure environment. The simulations focused on generating realistic cybersecurity events that could be monitored through Microsoft Sentinel and Log Analytics Workspace.

The testing process included:

- Brute force login attempts
- Authentication failure generation
- Port scanning activity using Nmap
- Log verification
- Alert validation
- Incident creation
- Dashboard monitoring
- Event correlation

The attack simulations were performed in a controlled environment to avoid affecting external systems or violating security policies. All testing was conducted within the cloud infrastructure created during implementation.

The primary objective of this chapter is to demonstrate that the proposed SOC environment can detect suspicious activities, store security logs, generate alerts, and provide analysts with sufficient visibility for investigation.

## **5.2 Brute Force Attack Simulation**

A brute force attack is one of the most common attack techniques used by attackers to gain unauthorized access to systems. This attack involves repeatedly attempting different passwords or credentials until access is granted.

In this project, a brute force simulation was performed against the Windows Server virtual machine to generate authentication failure logs.

### **Purpose of the Simulation**

The brute force simulation aimed to:

- Generate failed login events
- Test Security Event collection
- Validate Sentinel detection rules
- Produce alerts and incidents
- Verify monitoring visibility

### **Attack Scenario**

The simulation involved repeated login attempts using incorrect credentials.

The attacker perspective was simulated using repeated authentication requests.

### **Method Used**

The following approach was performed:

1. Connect to the public IP address of the virtual machine.
2. Attempt login using invalid credentials.
3. Repeat failed attempts multiple times.
4. Generate Event ID 4625.

5. Allow Azure Monitor Agent to forward logs.

### **Event Generation**

Windows Security Logs automatically generated failed authentication events.

Each failed login produced:

- Username attempted
- Timestamp
- Source IP
- Failure reason
- Event ID

### **Importance of Brute Force Simulation**

Brute force testing confirmed that the SOC environment can detect suspicious authentication behavior.

This attack scenario represents a realistic threat frequently encountered in enterprise environments.

## **5.3 Failed Login Event Analysis**

Failed login analysis focused on studying the events generated during brute force simulation.

Windows logs store failed authentication events using Security Event ID 4625.

These events are essential because they provide evidence of unauthorized access attempts.

### **Event ID 4625**

Event ID 4625 represents failed logon attempts.

The logs include several important attributes:

- Username
- Logon Type
- Source IP Address
- Failure Reason
- Time Generated
- Authentication Package

### **Query Used for Analysis**

The following Kusto Query Language (KQL) query was used:

```
SecurityEvent  
| where EventID == 4625  
| project TimeGenerated, Account, Computer, IpAddress  
| order by TimeGenerated desc
```

### **Analysis Purpose**

The analysis allowed investigators to:

- Identify repeated attempts
- Observe attacker behavior
- Monitor timestamps
- Detect login patterns

### **Findings**

The analysis demonstrated that authentication failures were correctly ingested into Log Analytics Workspace.

Repeated login attempts appeared chronologically.

This confirmed:

- Proper monitoring integration
- Functional event forwarding
- Successful DCR configuration

### **Security Importance**

Failed login analysis is important because attackers commonly attempt credential guessing.

SOC analysts rely on authentication monitoring to identify early-stage attacks.

## **5.4 Nmap Port Scanning Simulation**

Port scanning is commonly used by attackers during reconnaissance.

The objective of port scanning is to discover open ports and identify running services.

In this project, Nmap was used to simulate reconnaissance activity against the public IP of the virtual machine.

### **Purpose of Nmap Testing**

The simulation aimed to:

- Generate network activity
- Trigger logs
- Validate detection capability
- Test monitoring visibility

### **Tool Used**

The simulation used the tool:

□entity□["software","Nmap","Network scanning and reconnaissance tool"]□

### **Scan Example**

nmap -sS -p 1-1000 PUBLIC\_IP

### **Scan Behavior**

Nmap performed:

- TCP SYN scans
- Port discovery
- Service probing

### **Logs Generated**

Port scanning produced:

- Connection attempts
- Firewall logs
- Network monitoring events
- Windows Security activity

### **Detection Visibility**

The SOC environment successfully captured unusual network behavior.

Repeated connection attempts were observable inside logs.

## **Importance of Port Scanning Detection**

Reconnaissance is often the first stage of an attack lifecycle.

Detecting scanning activity allows analysts to identify suspicious actors before exploitation occurs.

## **5.5 Security Event Collection**

Security event collection refers to gathering all generated logs during testing.

The SOC environment collected logs from multiple sources.

### **Collected Event Categories**

The following logs were collected:

Authentication logs

Security events

- Windows Defender logs
- Sysmon logs
- Heartbeat monitoring logs
- Azure Activity Logs

### **Collection Workflow**

The log flow followed this path:

1. Event generated on VM.
2. Azure Monitor Agent collects data.
3. Data Collection Rule filters logs.
4. Logs sent to Log Analytics Workspace.
5. Sentinel processes events.

### **Verification Query**

SecurityEvent

| summarize count() by EventID

### **Collection Success Indicators**

Successful collection was confirmed when:

- Events appeared inside Log Analytics
- Dashboard updated
- Analytics rules executed

### **Importance of Security Collection**

Without reliable event collection, the SOC cannot detect or investigate threats.

Event collection is the backbone of SIEM functionality.

## **5.6 Detection Rules Validation**

Detection validation ensures that analytics rules function correctly.

The SOC used Sentinel scheduled analytics rules.

### **Validation Objective**

The purpose was to confirm:

- Rule execution
- Accurate alert creation
- Event matching
- Incident generation

### **Example Rule**

SecurityEvent

| where EventID == 4625

| summarize FailedAttempts=count() by Account, bin(TimeGenerated, 5m)

| where FailedAttempts > 5

### **Validation Process**

1. Generate failed logins.
2. Wait for ingestion.
3. Confirm rule execution.

4. Verify alert appearance.

## **Results**

Detection rules triggered successfully.

Alerts appeared after the threshold condition was met.

## **Importance**

Detection validation proves that monitoring logic works correctly.

Without rule validation, SOC analysts cannot trust detection accuracy.

## **5.7 Incident Generation**

Incident generation is the process of grouping alerts into manageable investigations.

Sentinel automatically creates incidents.

### **Purpose of Incidents**

Incidents help analysts:

- Investigate suspicious activity
- Track attack timeline
- Group alerts
- Prioritize threats

### **Incident Workflow**

1. Detection rule triggered.
2. Alert generated.
3. Sentinel creates incident.
4. Analyst investigates.

### **Incident Information**

Each incident includes:

- Severity level
- Related entities
- Timestamp
- Alert details

- Investigation graph

### **Validation Results**

Incidents appeared successfully after brute force simulation.

### **SOC Importance**

Incident generation improves operational efficiency.

It prevents analysts from manually reviewing every log.

## **5.8 Dashboard Monitoring Results**

Dashboards provide visual confirmation of attack activity.

The Sentinel workbook displayed monitoring results.

### **Dashboard Components Used**

The dashboard included:

- Failed login counters
- Timeline graphs
- Event statistics
- Top IP addresses
- Security activity tables

### **Observed Results**

The dashboard reflected:

- Increased authentication failures
- Attack timestamps
- Event trends
- Detection activity

### **Visualization Benefits**

Dashboards simplify interpretation.

SOC analysts can quickly identify suspicious patterns.

### **Testing Outcome**

The dashboard updated automatically.

This confirmed:

- Real-time ingestion
- Workbook functionality
- Proper data visualization

### **Security Advantage**

Visual monitoring accelerates threat awareness.

## **5.9 Attack Correlation**

Attack correlation refers to linking multiple events together.

Rather than analyzing logs independently, SOC systems identify relationships between activities.

### **Correlation Purpose**

The objective was to connect:

- Failed logins
- Network scans
- Repeated IP activity
- Multiple timestamps

### **Example Correlation Scenario**

1. Nmap scan detected.
2. Failed login attempts follow.
3. Same IP appears repeatedly.
4. Sentinel links activity.

### **Benefits**

Correlation improves:

- Context awareness
- Attack understanding
- Threat prioritization

### **SOC Intelligence**

Correlation reduces false positives.

Multiple related events indicate stronger evidence of malicious behavior.

### **Importance**

Attack correlation transforms isolated logs into meaningful security incidents.

## **5.10 Testing Methodology**

Testing methodology describes the structured approach used to evaluate system performance.

The methodology followed a controlled cybersecurity testing model.

### **Testing Phases**

The project followed several stages:

#### **Phase 1: Environment Validation**

Confirm infrastructure readiness.

#### **Phase 2: Agent Verification**

Verify Azure Monitor Agent communication.

#### **Phase 3: Log Collection Validation**

Confirm logs reach Log Analytics.

#### **Phase 4: Attack Simulation**

Perform brute force and Nmap testing.

#### **Phase 5: Detection Validation**

Ensure alerts appear.

#### **Phase 6: Incident Validation**

Confirm incident creation.

#### **Phase 7: Dashboard Analysis**

Review visual results.

### **Testing Tools Used**

- PowerShell
- Remote Desktop
- Windows Event Viewer
- Nmap
- Log Analytics Workspace
- Microsoft Sentinel

## **CHAPTER 6 — RESULTS AND DISCUSSION**

### **6.1 Introduction**

The results and discussion chapter presents the outcomes obtained after implementing and testing the cloud-based Security Operations Center (SOC). This chapter analyzes the effectiveness of the deployed environment by examining collected logs, attack detection capabilities, incident generation, dashboard visualization, system performance, and overall threat monitoring efficiency.

The purpose of this chapter is to evaluate whether the project objectives were successfully achieved. Since the project focuses on designing and implementing a cloud-based SOC using Microsoft Azure and Microsoft Sentinel, measurable outcomes are required to demonstrate system functionality.

The testing phase produced a large number of security events generated from simulated attacks, monitoring agents, endpoint protection systems, and cloud activity logs. These events provided valuable data that allowed the system to demonstrate its monitoring and analytical capabilities.

The discussion section also explains how different components interacted with each other during attack scenarios. It evaluates the effectiveness of security monitoring workflows and highlights both strengths and limitations.

The results obtained from this project demonstrate that a cloud-based SOC can provide centralized visibility, automated detection, event monitoring, and incident management.

This chapter provides detailed interpretation of the collected results to support the validity of the proposed architecture.

### **6.2 Log Collection Results**

Log collection is one of the most critical functions of a Security Operations Center. The SOC environment depends entirely on logs to identify suspicious behavior, monitor activities, and generate alerts.

During implementation and testing, the Azure environment successfully collected logs from multiple sources.

### **Log Sources Used**

The following sources generated data:

- Windows Security Logs
- Sysmon Operational Logs
- Windows Defender Logs
- Azure Activity Logs
- Azure Monitor Agent Heartbeat Logs
- Network Activity Logs

### **Log Collection Workflow**

The logs followed a structured ingestion process:

1. Event generated on the virtual machine.
2. Azure Monitor Agent captured the event.
3. Data Collection Rules filtered the data.
4. Logs transferred to Log Analytics Workspace.
5. Sentinel analyzed the logs.

### **Log Volume Observed**

The system generated a substantial number of events during testing.

Examples included:

- SecurityEvent table entries
- Heartbeat monitoring events
- Authentication failure records
- Endpoint activity logs

### **Query Used for Validation**

SecurityEvent

| summarize Count = count() by EventID

| order by Count desc

### **Result Interpretation**

The query confirmed that events were stored successfully.

The most frequently observed event IDs included:

- Event ID 4624 (Successful Login)
- Event ID 4625 (Failed Login)
- Event ID 4688 (Process Creation)
- Event ID 5156 (Network Filtering Platform)

### **Collection Reliability**

No major data loss was observed.

Heartbeat monitoring indicated stable agent communication.

### **Importance of Results**

The successful collection of logs confirmed:

- Proper Azure Monitor Agent deployment
- Functional Data Collection Rules
- Stable connectivity
- Successful workspace integration

Reliable log collection proved that the monitoring environment was functioning correctly.

## **6.3 Attack Detection Results**

Attack detection is one of the most important measurements of SOC effectiveness.

The testing process included simulated brute force attacks and network reconnaissance using port scanning.

The SOC successfully detected suspicious behavior.

### **Brute Force Detection Results**

During repeated login attempts, Sentinel identified authentication failures.

The detection rule monitored:

Event ID 4625

- Repeated failed attempts
- Short time intervals

### **Alert Triggering**

Analytics rules generated alerts when conditions were met.

### **Detection Timing**

Detection occurred shortly after logs were ingested.

### **Nmap Detection Results**

Port scanning activity produced unusual network patterns.

The SOC identified:

- Repeated connection attempts
- High-volume port access
- Suspicious network activity

### **Query Used**

SecurityEvent

| where EventID == 5156

| summarize count() by IpAddress

### **Detection Visibility**

Both attack scenarios appeared inside Sentinel.

### **Detection Success**

The system successfully recognized:

- Unauthorized authentication attempts
- Reconnaissance activity
- Suspicious behavior patterns

### **Result Evaluation**

Detection capability demonstrated that the SOC environment could identify malicious actions in near real time.

## **6.4 Incident Response Results**

Incident response is a core function of modern SOC environments.

When suspicious behavior is detected, Sentinel creates incidents that allow analysts to investigate and respond.

### **Incident Creation Results**

Incidents were generated automatically.

### **Incident Workflow**

1. Attack occurred.
2. Logs generated.
3. Analytics rule triggered.
4. Alert created.
5. Incident generated.

### **Incident Details Observed**

Each incident included:

- Severity level
- Related entities
- Attack timeline
- Triggered rule
- Timestamp

### **Example Incident**

Brute force activity generated a medium-severity incident.

### **Analyst Visibility**

The incident interface allowed investigation.

### **Response Efficiency**

The SOC environment reduced manual monitoring.

### **Incident Benefits**

- Faster investigation
- Centralized alert grouping
- Reduced analysis time

## **Validation**

Incidents were verified inside the Incidents section of Microsoft Sentinel.

## **Result Conclusion**

Incident response capabilities confirmed that Sentinel successfully converted alerts into actionable investigations.

## **6.5 Dashboard Visualization Results**

Visualization plays an essential role in SOC operations because analysts require a quick understanding of ongoing activity.

Dashboards created using Sentinel Workbooks provided real-time visibility.

## **Dashboard Components**

The workbook displayed:

- Total failed login attempts
- Timeline charts
- Event frequency
- Top source IPs
- Event category statistics
- Security monitoring summaries

## **Visualization Effectiveness**

The dashboard updated automatically.

## **Monitoring Visibility**

Attack activity became visible in graphical format.

## **Results Observed**

During brute force testing:

- Failed login counts increased

- Charts showed event spikes
- Activity patterns became visible

### **Geographic Visualization**

Map-based monitoring displayed event sources.

### **Dashboard Benefits**

- Faster analysis
- Improved understanding
- Better visibility
- Simplified investigation

### **Workbook Role**

Workbooks provided an analyst-friendly interface.

### **Result Evaluation**

Visualization confirmed that security monitoring could be interpreted efficiently.

## **6.6 System Performance Analysis**

System performance measures how efficiently the SOC environment processes data.

Performance analysis focused on:

- Data ingestion speed
- Query execution time
- Dashboard responsiveness
- Alert generation delay

### **Data Ingestion Performance**

Logs were transferred quickly from the virtual machine.

### **Query Performance**

Kusto Query Language queries executed efficiently.

### **Monitoring Delay**

Minor delay existed between event generation and dashboard visibility.

### **System Stability**

The Azure environment remained stable.

### **Resource Consumption**

The environment consumed moderate cloud resources.

### **Performance Factors**

Performance depended on:

- Internet connectivity
- Azure processing
- Log volume
- Query complexity

### **Observed Efficiency**

The monitoring platform remained responsive.

### **Result Interpretation**

Performance analysis confirmed that the cloud-based SOC environment operated effectively.

## **6.7 Threat Detection Accuracy**

Threat detection accuracy refers to the ability of the system to identify malicious behavior correctly.

A highly accurate SOC should detect attacks while minimizing false positives.

### **Accuracy Evaluation Method**

The project compared generated attacks against detection results.

### **Testing Criteria**

The system was evaluated using:

- Attack presence
- Alert generation
- Incident creation
- Event correlation

## **Detection Accuracy Results**

The SOC successfully identified:

- Failed logins
- Repeated authentication attempts
- Network scanning
- Suspicious patterns

## **False Positives**

Few irrelevant alerts were observed.

## **Detection Reliability**

The analytics rules provided consistent performance.

## **SOC Accuracy Benefits**

High detection accuracy improves:

- Analyst confidence
- Faster response
- Reduced noise

## **Result Interpretation**

The project demonstrated strong detection capability.

## **Importance**

Threat detection accuracy is critical because inaccurate monitoring reduces SOC effectiveness.

## **6.8 Discussion**

The results obtained from this project demonstrate that cloud-native security monitoring can provide strong visibility into system activity.

The implemented SOC successfully achieved its primary objectives.

## **Achievements Observed**

The project successfully:

- Collected logs centrally
- Detected suspicious activity

- Generated alerts
- Created incidents
- Visualized attacks
- Supported investigation

### **Azure Benefits**

Microsoft Azure provided an integrated environment.

### **Sentinel Advantages**

Microsoft Sentinel simplified SIEM deployment.

### **Monitoring Strengths**

The SOC environment demonstrated:

- Strong scalability
- Cloud-native monitoring
- Centralized visibility
- Automated workflows

### **Practical Importance**

The project showed that enterprise-grade monitoring can be achieved using cloud services.

### **Challenges Identified**

Several challenges were observed:

- Learning curve of Azure services
- Query complexity
- Log ingestion timing
- Connector configuration

### **Educational Value**

The project provided hands-on experience.

### **Overall Discussion**

The implementation demonstrated that a cloud-based SOC can function effectively within an academic environment while providing realistic cybersecurity monitoring capabilities.

## **CHAPTER 7 — CONCLUSION AND FUTURE WORK**

### **7.1 Conclusion**

The conclusion chapter summarizes the overall achievements of the project and evaluates the effectiveness of the implemented cloud-based Security Operations Center (SOC). Throughout this project, a complete cybersecurity monitoring environment was designed, deployed, configured, and tested using cloud-native technologies within Microsoft Azure.

The primary objective of this project was to build a centralized monitoring system capable of collecting logs, detecting suspicious activity, generating alerts, visualizing attacks, and supporting incident investigation. The implementation successfully demonstrated that modern SOC functionality can be achieved using cloud services without requiring expensive physical infrastructure.

The project combined several critical cybersecurity technologies including Microsoft Sentinel, Log Analytics Workspace, Azure Monitor Agent, Windows Defender, Sysmon, dashboards, analytics rules, and automation workflows.

During implementation, a virtual machine was deployed as the monitored endpoint. Logs generated by authentication events, system activity, and network behavior were successfully collected and transmitted to the monitoring environment.

The SOC environment proved capable of:

- Collecting security logs
- Monitoring suspicious activity
- Detecting attacks
- Generating incidents
- Supporting investigation
- Visualizing attack data

Attack simulation demonstrated the practical capability of the environment.

Brute force attempts and Nmap port scans successfully triggered detection logic, proving that the SOC could identify real-world attack behaviors.

The results confirmed that cloud-native SIEM solutions provide a strong alternative to traditional security monitoring systems.

This project also highlighted the importance of integrating multiple monitoring components together. A SOC does not rely on a single technology but instead depends on collaboration between monitoring agents, data collection rules, analytics engines, dashboards, and security workflows.

The overall project successfully fulfilled its intended purpose and demonstrated how a cloud-based SOC can provide enterprise-style monitoring within an educational environment.

## **7.2 Achieved Objectives**

The project successfully achieved the major objectives defined during the planning and analysis phases.

The completed environment provided practical cybersecurity monitoring capabilities while remaining aligned with the academic scope of the project.

### **Objective 1: Deploy Cloud Infrastructure**

A secure cloud environment was deployed using Microsoft Azure.

### **Objective 2: Create Monitoring Endpoint**

A Windows Server virtual machine was successfully created.

### **Objective 3: Enable Log Collection**

Azure Monitor Agent and Data Collection Rules successfully collected events.

### **Objective 4: Integrate Microsoft Sentinel**

Sentinel was enabled and connected to Log Analytics Workspace.

### **Objective 5: Generate Security Visibility**

Logs appeared inside dashboards and monitoring queries.

### **Objective 6: Simulate Attacks**

Brute force and Nmap attacks were performed.

### **Objective 7: Validate Detection Rules**

Analytics rules triggered alerts successfully.

**Objective 8: Generate Incidents**

Incidents appeared automatically after suspicious activity.

**Objective 9: Create Visualization Dashboards**

Workbooks displayed attack statistics and monitoring trends.

**Objective 10: Implement Automation**

Logic Apps automation demonstrated response workflows.

These achievements confirmed that the proposed architecture functioned successfully.

**7.3 Project Contributions**

This project contributes to cybersecurity education and cloud monitoring research by demonstrating how a Security Operations Center can be implemented using accessible cloud technologies.

Several contributions were achieved.

**Contribution to Academic Learning**

The project provides a practical example of SOC deployment.

**Contribution to Cloud Security Research**

The implementation demonstrates how cloud-native monitoring can replace traditional infrastructure.

**Contribution to SIEM Understanding**

The project explains how SIEM technology operates.

**Contribution to Threat Detection**

Attack simulations provide realistic examples of security monitoring.

**Contribution to Azure-Based Security**

The project demonstrates integration between Azure services.

**Contribution to Incident Response**

The implementation illustrates incident generation.

**Contribution to Visualization**

Dashboards improve interpretation of security events.

**Contribution to Cybersecurity Awareness**

The project highlights the importance of proactive monitoring.

### **Contribution to Educational Projects**

The project provides a reproducible architecture.

### **Contribution to Security Operations**

The environment demonstrates SOC workflow.

The project contributes both theoretically and practically.

## **7.4 Challenges Faced**

Several challenges were encountered during project implementation.

These challenges provided valuable learning opportunities.

### **Azure Service Complexity**

Microsoft Azure includes many interconnected services.

Understanding relationships between services required time.

### **Data Collection Rules Configuration**

DCR configuration was initially difficult.

Incorrect settings prevented log collection.

### **Log Visibility Delays**

Some logs required time before appearing.

### **Connector Configuration**

Some connectors required additional setup.

### **Map Visualization Issues**

Geographic mapping required proper IP enrichment.

### **Query Development**

Kusto Query Language required practice.

### **Incident Timing**

Analytics rules execute based on schedule.

### **Networking Challenges**

NSG configuration required careful planning.

## **Cloud Learning Curve**

Azure architecture required technical understanding.

## **Resource Constraints**

Azure Student Subscription imposed limitations.

Despite these challenges, all major obstacles were successfully resolved.

## **7.5 Recommendations**

Several recommendations can improve the SOC environment and increase monitoring effectiveness.

### **Expand Log Sources**

Future implementations should integrate more endpoints.

### **Add IDS/IPS Integration**

Network detection tools can improve visibility.

Examples include:

- Suricata
- Snort

### **Implement Threat Intelligence**

Threat feeds improve IP reputation analysis.

### **Improve Automation**

Additional workflows can reduce analyst workload.

### **Add Multi-Endpoint Monitoring**

Multiple virtual machines provide realistic monitoring.

### **Integrate Honeypot Systems**

Honeypots generate additional attack visibility.

### **Improve Detection Logic**

More analytics rules improve coverage.

### **Enhance Dashboard Design**

Additional charts improve interpretation.

### **Add AI Integration**

AI can summarize incidents.

### **Enable Continuous Monitoring**

Continuous improvement enhances SOC maturity.

These recommendations would strengthen future deployments.

## **7.6 Future Work**

Future work refers to improvements that could be added beyond the current implementation.

Although the project achieved its objectives, several advanced features can enhance the environment.

### **Multi-VM Deployment**

Future environments could monitor multiple servers.

### **Full Enterprise Architecture**

Expand SOC coverage across departments.

### **IDS/IPS Integration**

Integrate intrusion detection systems.

### **Threat Intelligence Integration**

Connect external feeds.

### **AI-Based Log Analysis**

Integrate Azure OpenAI.

### **Automated Remediation**

Automatically block suspicious IPs.

### **Cloud-Native Honeypots**

Deploy honeypots.

### **Machine Learning Detection**

Behavioral analytics can identify anomalies.

### **Cross-Cloud Monitoring**

Monitor multiple cloud providers.

## **Advanced Incident Response**

Add playbooks.

## **Real-Time Threat Hunting**

Expand hunting capabilities.

# **APPENDICES**

## **Appendix A: KQL Queries**

This appendix contains the Kusto Query Language (KQL) queries used throughout the project. These queries were applied within Microsoft Sentinel and Log Analytics Workspace to analyze security events, monitor suspicious activity, generate dashboards, and validate attack simulations.

### **A.1 Failed Login Detection Query**

Purpose: Detect repeated authentication failures.

SecurityEvent

| where EventID == 4625

| summarize FailedAttempts = count() by Account, bin(TimeGenerated, 5m)

| where FailedAttempts > 5

Explanation:

This query searches for failed login events and identifies accounts with multiple failed login attempts within a short time period.

### **A.2 Successful Login Query**

Purpose: Display successful user authentications.

SecurityEvent

| where EventID == 4624

| project TimeGenerated, Account, IpAddress, Computer

| order by TimeGenerated desc

Explanation:

This query retrieves successful login activity.

### **A.3 Top Attacking IP Addresses**

Purpose: Identify IP addresses generating the highest number of suspicious events.

SecurityEvent

| where EventID == 4625

| summarize Attempts = count() by IpAddress

| top 10 by Attempts desc

Explanation:

This query identifies the top source IP addresses responsible for failed login attempts.

#### **A.4 Security Event Distribution**

Purpose: Display distribution of event IDs.

SecurityEvent

| summarize EventCount = count() by EventID

| order by EventCount desc

Explanation:

This query summarizes event frequency.

#### **A.5 Heartbeat Validation Query**

Purpose: Confirm Azure Monitor Agent communication.

Heartbeat

| summarize LastSeen=max(TimeGenerated) by Computer

Explanation:

Heartbeat logs confirm endpoint connectivity.

#### **A.6 Process Creation Monitoring**

Purpose: Detect new process execution.

SecurityEvent

| where EventID == 4688

| project TimeGenerated, Computer, Account, Process

Explanation:

Tracks process creation activity.

#### **A.7 Network Activity Monitoring**

Purpose: Monitor network filtering platform events.

SecurityEvent

| where EventID == 5156

| summarize count() by IPAddress

Explanation:

Displays allowed network connections.

### **A.8 Log Collection Verification**

Purpose: Verify ingestion into Log Analytics Workspace.

SecurityEvent

| take 50

Explanation:

Displays recent security events.

---

## **Appendix B: Analytics Rules**

This appendix contains analytics rules configured within Microsoft Sentinel.

### **B.1 Brute Force Detection Rule**

Rule Name: Brute Force Login Detection

Description:

Detect repeated failed login attempts.

Severity: Medium

Query:

SecurityEvent

| where EventID == 4625

| summarize FailedAttempts=count() by Account, bin(TimeGenerated, 5m)

| where FailedAttempts > 5

Trigger Threshold:

5 failed logins within 5 minutes.

MITRE ATT&CK Mapping:

- Credential Access
- Brute Force

## **B.2 Port Scan Detection Rule**

Rule Name: Network Reconnaissance Detection

Description:

Detect repeated connection attempts.

Severity: Medium

Query:

SecurityEvent

| where EventID == 5156

| summarize Attempts=count() by IpAddress, bin(TimeGenerated, 5m)

| where Attempts > 100

Trigger Threshold:

High-volume network activity.

MITRE ATT&CK Mapping:

- Discovery
- Network Service Scanning

## **B.3 Suspicious Process Monitoring**

Rule Name: Unusual Process Execution

Severity: Low

Query:

SecurityEvent

| where EventID == 4688

| where Process contains "powershell"

## **B.4 Heartbeat Missing Detection**

Rule Name: Agent Communication Failure

Severity: High

Query:

Heartbeat

| summarize LastSeen=max(TimeGenerated) by Computer

| where LastSeen < ago(30m)

---

## Appendix C: Sysmon Configuration

This appendix describes Sysmon deployment and configuration.

### C.1 Sysmon Installation Command

Sysmon64.exe -accepteula -i sysmonconfig.xml

### C.2 Sysmon Purpose

Sysmon provides advanced monitoring.

It records:

- Process creation
- Network connections
- Registry modifications
- Driver loading
- DNS queries
- File creation

### C.3 Sysmon XML Configuration Example

```
<Sysmon schemaversion="4.90">
```

```
<EventFiltering>
```

```
<ProcessCreate onmatch="include" />
```

```
<NetworkConnect onmatch="include" />
```

<RegistryEvent onmatch="include" />

<FileCreate onmatch="include" />

</EventFiltering>

</Sysmon>

#### C.4 Sysmon Event IDs

| Event ID | Description        |
|----------|--------------------|
| 1        | Process Creation   |
| 3        | Network Connection |
| 7        | Image Loaded       |
| 11       | File Created       |
| 13       | Registry Value Set |
| 22       | DNS Query          |

#### C.5 Sysmon Integration Workflow

1. Sysmon installed on Windows Server.
2. Events generated locally.
3. Azure Monitor Agent collects logs.
4. Data sent to Log Analytics Workspace.
5. Sentinel analyzes activity.

# Appendix D: Screenshots

This appendix contains screenshots captured during implementation and testing.

Figure D.1 — Azure Resource Group

Screenshot showing the Resource Group containing project components.

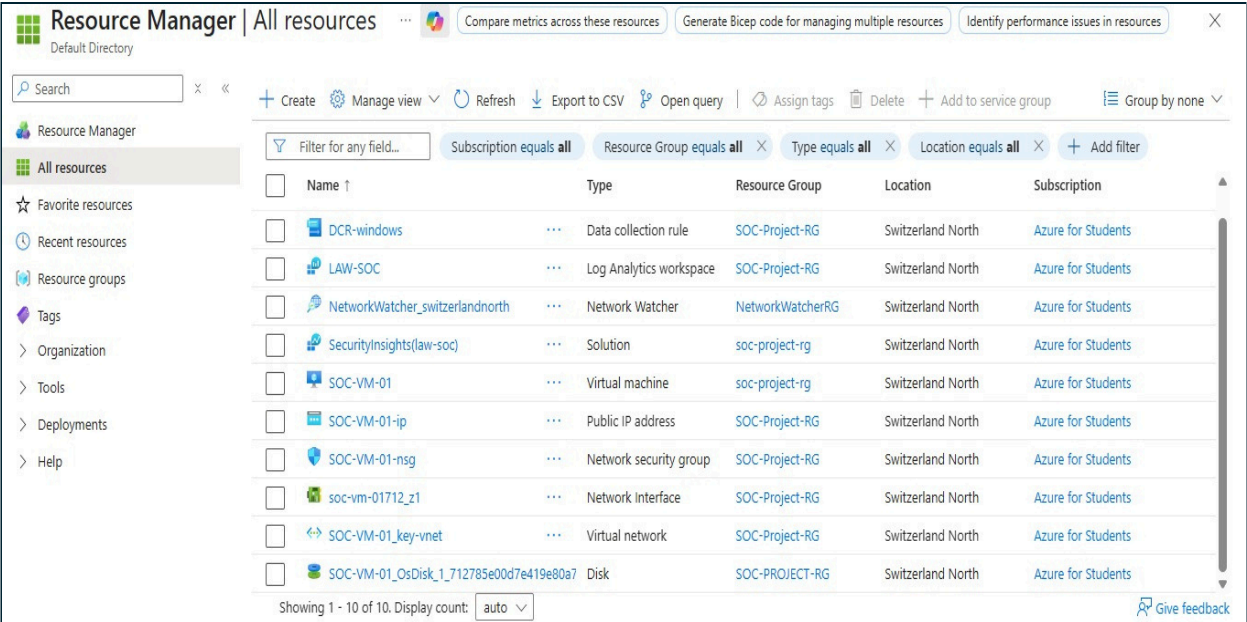


Figure D.2 — Windows Server Virtual Machine

Screenshot displaying VM configuration.

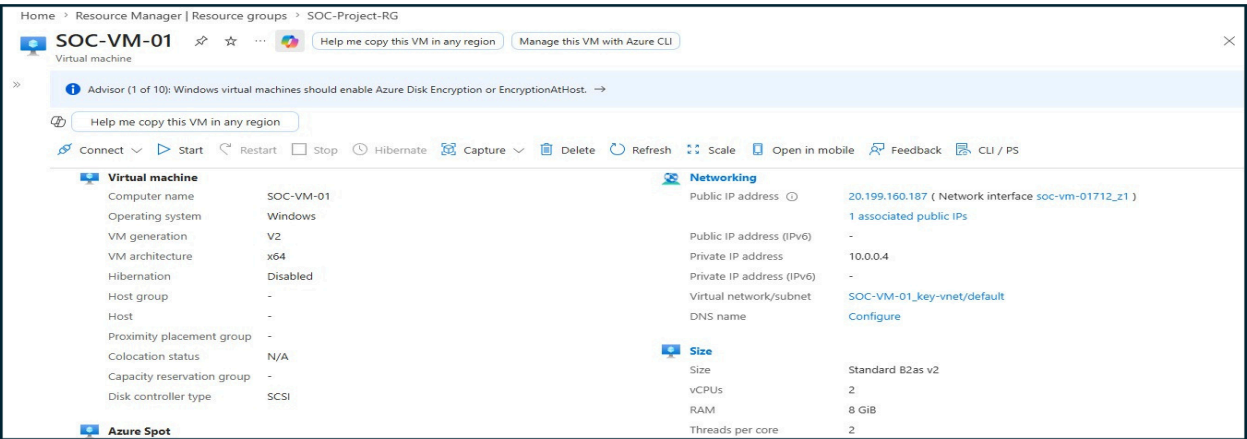


Figure D.3 — Public Static IP Configuration

Screenshot showing assigned public IP.

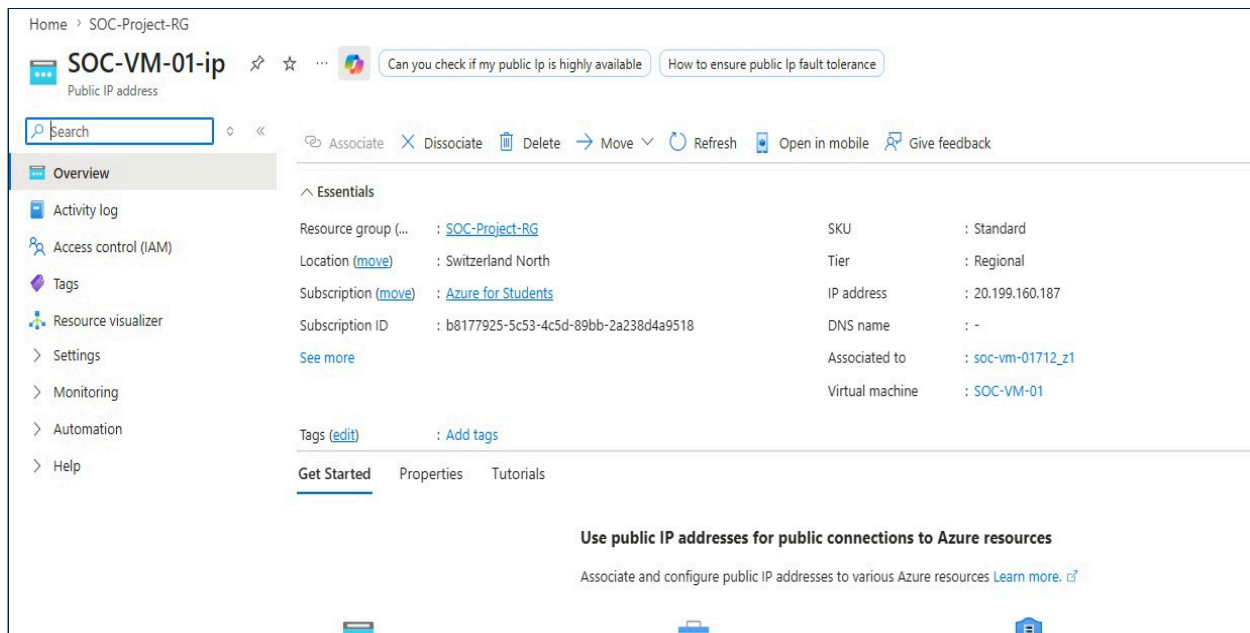


Figure D.4 — Azure Virtual Network

Screenshot of VNet configuration.

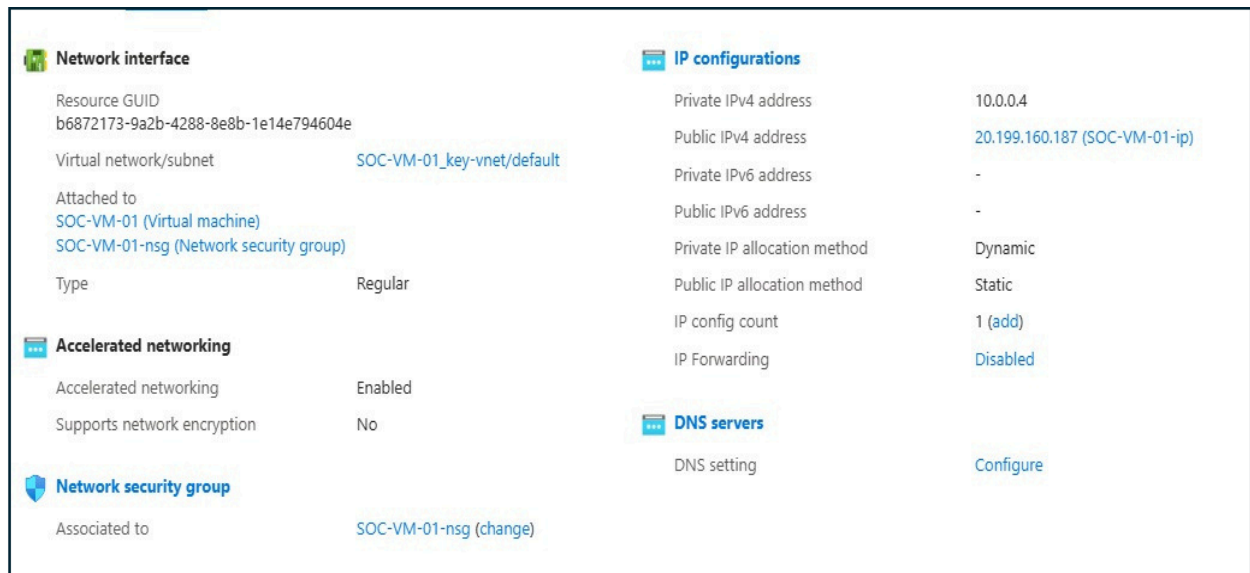


Figure D.5 — Network Security Group Rules

Screenshot of inbound and outbound rules.

The screenshot displays the Windows Defender Firewall with Advanced Security console. The left sidebar shows the navigation pane with 'Inbound Rules' and 'Outbound Rules' selected. The main pane shows a list of rules with columns: Name, Group, Profile, Enabled, Action, Override, Program, Local Address, Remote Address, Protocol, and Local Port. The right sidebar shows the 'Actions' pane with options like 'New Rule', 'Filter...', 'Filter...', 'View', 'Refresh', 'Export', and 'Help'.

| Name                                     | Group                        | Profile   | Enabled | Action | Override | Program     | Local Address | Remote Address | Protocol | Local Port    |
|------------------------------------------|------------------------------|-----------|---------|--------|----------|-------------|---------------|----------------|----------|---------------|
| iSCSI Service (TCP-In)                   | iSCSI Service                | All       | No      | Allow  | No       | %System...  | Any           | Any            | TCP      | Any           |
| Key Management Service (TCP-In)          | Key Management Service       | All       | No      | Allow  | No       | %System...  | Any           | Any            | TCP      | 1688          |
| mDNS (UDP-In)                            | mDNS                         | Public    | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 5353          |
| mDNS (UDP-In)                            | mDNS                         | Private   | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 5353          |
| mDNS (UDP-In)                            | mDNS                         | Domain    | Yes     | Allow  | No       | %System...  | Any           | Any            | UDP      | 5353          |
| Microsoft Edge (mDNS-In)                 | Microsoft Edge               | All       | Yes     | Allow  | No       | C:\Progr... | Any           | Any            | UDP      | 5353          |
| Microsoft Media Foundation Network So... | Microsoft Media Foundatio... | All       | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | TCP      | 554, 8554-... |
| Microsoft Media Foundation Network So... | Microsoft Media Foundatio... | All       | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 5000-5020     |
| Netlogon Service (NP-In)                 | Netlogon Service             | All       | No      | Allow  | No       | System      | Any           | Any            | TCP      | 445           |
| Netlogon Service Authz (RPC)             | Netlogon Service             | All       | No      | Allow  | No       | %System...  | Any           | Any            | TCP      | RPC Dyna...   |
| Network Discovery (LLMNR-UDP-In)         | Network Discovery            | Private   | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 5355          |
| Network Discovery (LLMNR-UDP-In)         | Network Discovery            | Domain... | No      | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 5355          |
| Network Discovery (NB-Datagram-In)       | Network Discovery            | Domain... | No      | Allow  | No       | System      | Any           | Any            | UDP      | 138           |
| Network Discovery (NB-Datagram-In)       | Network Discovery            | Private   | Yes     | Allow  | No       | System      | Any           | Any            | UDP      | 138           |
| Network Discovery (NB-Name-In)           | Network Discovery            | Domain... | No      | Allow  | No       | System      | Any           | Any            | UDP      | 137           |
| Network Discovery (NB-Name-In)           | Network Discovery            | Private   | Yes     | Allow  | No       | System      | Any           | Any            | UDP      | 137           |
| Network Discovery (Pub-WSD-In)           | Network Discovery            | Private   | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 3702          |
| Network Discovery (Pub-WSD-In)           | Network Discovery            | Domain... | No      | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 3702          |
| Network Discovery (SSDP-In)              | Network Discovery            | Private   | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 1900          |
| Network Discovery (SSDP-In)              | Network Discovery            | Domain... | No      | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 1900          |
| Network Discovery (UPnP-In)              | Network Discovery            | Domain... | No      | Allow  | No       | System      | Any           | Any            | TCP      | 2869          |
| Network Discovery (UPnP-In)              | Network Discovery            | Private   | Yes     | Allow  | No       | System      | Any           | Any            | TCP      | 2869          |
| Network Discovery (WSD Events-In)        | Network Discovery            | Private   | Yes     | Allow  | No       | System      | Any           | Any            | TCP      | 5357          |
| Network Discovery (WSD Events-In)        | Network Discovery            | Domain... | No      | Allow  | No       | System      | Any           | Any            | TCP      | 5357          |
| Network Discovery (WSD EventsSecure-In)  | Network Discovery            | Domain... | No      | Allow  | No       | System      | Any           | Any            | TCP      | 5358          |
| Network Discovery (WSD EventsSecure-In)  | Network Discovery            | Private   | Yes     | Allow  | No       | System      | Any           | Any            | TCP      | 5358          |
| Network Discovery (WSD-In)               | Network Discovery            | Domain... | No      | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 3702          |
| Network Discovery (WSD-In)               | Network Discovery            | Private   | Yes     | Allow  | No       | %System...  | Any           | Local subnet   | UDP      | 3702          |

| Name                                         | Group                         | Profile    | Enabled | Action | Override | Program    | Local Address | Remote Address   | Protocol | Local Port |
|----------------------------------------------|-------------------------------|------------|---------|--------|----------|------------|---------------|------------------|----------|------------|
| AllJoyn Router (TCP-Out)                     | AllJoyn Router                | Domain...  | Yes     | Allow  | No       | %System... | Any           | Any              | TCP      | Any        |
| AllJoyn Router (UDP-Out)                     | AllJoyn Router                | Domain...  | Yes     | Allow  | No       | %System... | Any           | Any              | UDP      | Any        |
| BranchCache Content Retrieval (HTTP-O...     | BranchCache - Content Retr... | All        | No      | Allow  | No       | SYSTEM     | Any           | Any              | TCP      | Any        |
| BranchCache Hosted Cache Client (HTTP...     | BranchCache - Hosted Cach...  | All        | No      | Allow  | No       | SYSTEM     | Any           | Any              | TCP      | Any        |
| BranchCache Hosted Cache Server (HTTP...     | BranchCache - Hosted Cach...  | All        | No      | Allow  | No       | SYSTEM     | Any           | Any              | TCP      | 80, 443    |
| BranchCache Peer Discovery (WSD-Out)         | BranchCache - Peer Discove... | All        | No      | Allow  | No       | %system... | Any           | Local subnet     | UDP      | Any        |
| Captive Portal Flow                          | Captive Portal Flow           | All        | Yes     | Allow  | No       | Any        | Any           | Any              | Any      | Any        |
| Captive Portal Flow                          | Captive Portal Flow           | All        | Yes     | Allow  | No       | Any        | Any           | Any              | Any      | Any        |
| Cast to Device functionality (qWave-TCP...   | Cast to Device functionality  | Private... | Yes     | Allow  | No       | %System... | Any           | PlayTo Renderers | TCP      | Any        |
| Cast to Device functionality (qWave-UDP...   | Cast to Device functionality  | Private... | Yes     | Allow  | No       | %System... | Any           | PlayTo Renderers | UDP      | Any        |
| Cast to Device streaming server (RTP-Stre... | Cast to Device functionality  | Domain     | Yes     | Allow  | No       | %System... | Any           | Any              | UDP      | Any        |
| Cast to Device streaming server (RTP-Stre... | Cast to Device functionality  | Private    | Yes     | Allow  | No       | %System... | Any           | Local subnet     | UDP      | Any        |
| Cast to Device streaming server (RTP-Stre... | Cast to Device functionality  | Public     | Yes     | Allow  | No       | %System... | Any           | PlayTo Renderers | UDP      | Any        |
| Core Networking - DNS (UDP-Out)              | Core Networking               | All        | Yes     | Allow  | No       | %System... | Any           | Any              | UDP      | Any        |
| Core Networking - Dynamic Host Config...     | Core Networking               | All        | Yes     | Allow  | No       | %System... | Any           | Any              | UDP      | 68         |
| Core Networking - Dynamic Host Config...     | Core Networking               | All        | Yes     | Allow  | No       | %System... | Any           | Any              | UDP      | 546        |
| Core Networking - Group Policy (LSASS...     | Core Networking               | Domain     | Yes     | Allow  | No       | %System... | Any           | Any              | TCP      | Any        |
| Core Networking - Group Policy (INP-Out)     | Core Networking               | Domain     | Yes     | Allow  | No       | System     | Any           | Any              | TCP      | Any        |
| Core Networking - Group Policy (TCP-Out)     | Core Networking               | Domain     | Yes     | Allow  | No       | %System... | Any           | Any              | TCP      | Any        |
| Core Networking - Internet Group Mana...     | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Any              | IGMP     | Any        |
| Core Networking - IPHTTPS (TCP-Out)          | Core Networking               | All        | Yes     | Allow  | No       | %System... | Any           | Any              | TCP      | Any        |
| Core Networking - IPv6 (IPv6-Out)            | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Any              | IPv6     | Any        |
| Core Networking - Multicast Listener Do...   | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Local subnet     | ICMPv6   | Any        |
| Core Networking - Multicast Listener Qu...   | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Local subnet     | ICMPv6   | Any        |
| Core Networking - Multicast Listener Rep...  | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Local subnet     | ICMPv6   | Any        |
| Core Networking - Multicast Listener Rep...  | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Local subnet     | ICMPv6   | Any        |
| Core Networking - Neighbor Discovery A...    | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Any              | ICMPv6   | Any        |
| Core Networking - Neighbor Discovery S...    | Core Networking               | All        | Yes     | Allow  | No       | System     | Any           | Any              | ICMPv6   | Any        |

Figure D.6 — Azure Monitor Agent Installation

Screenshot showing AMA configuration.

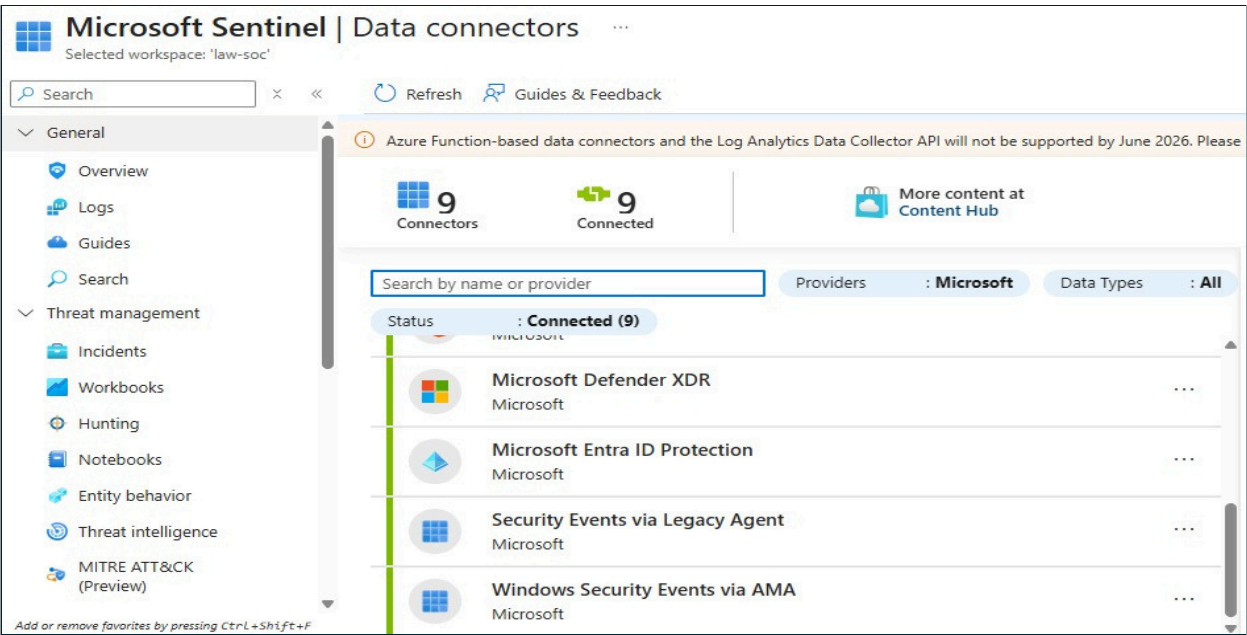


Figure D.7 — Log Analytics Workspace

Screenshot displaying workspace overview.

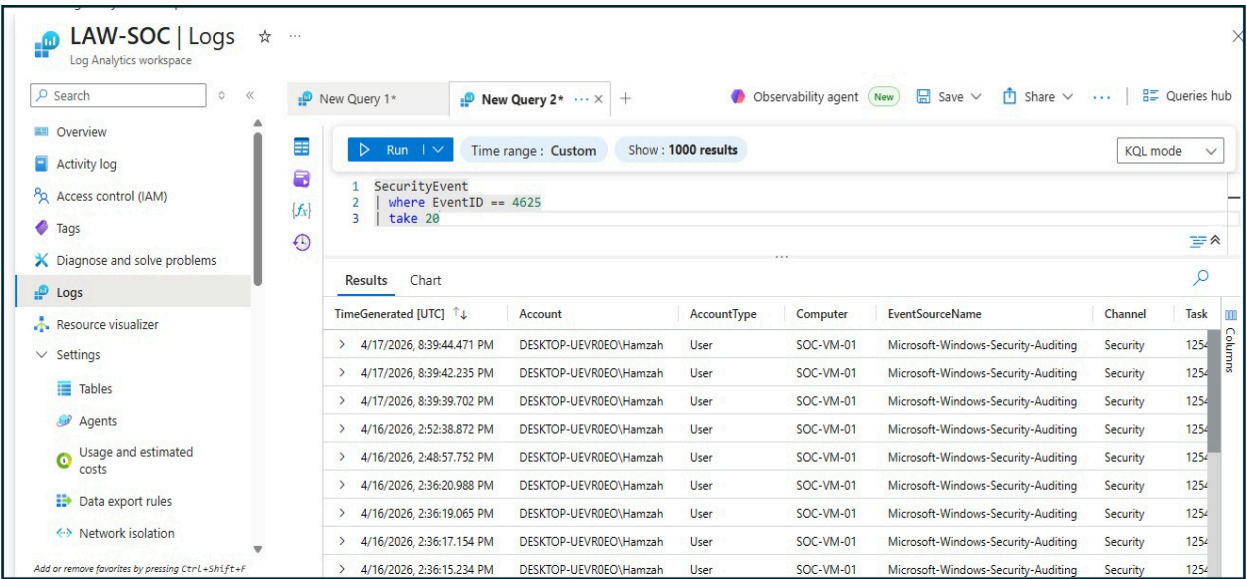


Figure D.8 — Microsoft Sentinel Overview

Screenshot of Sentinel dashboard.

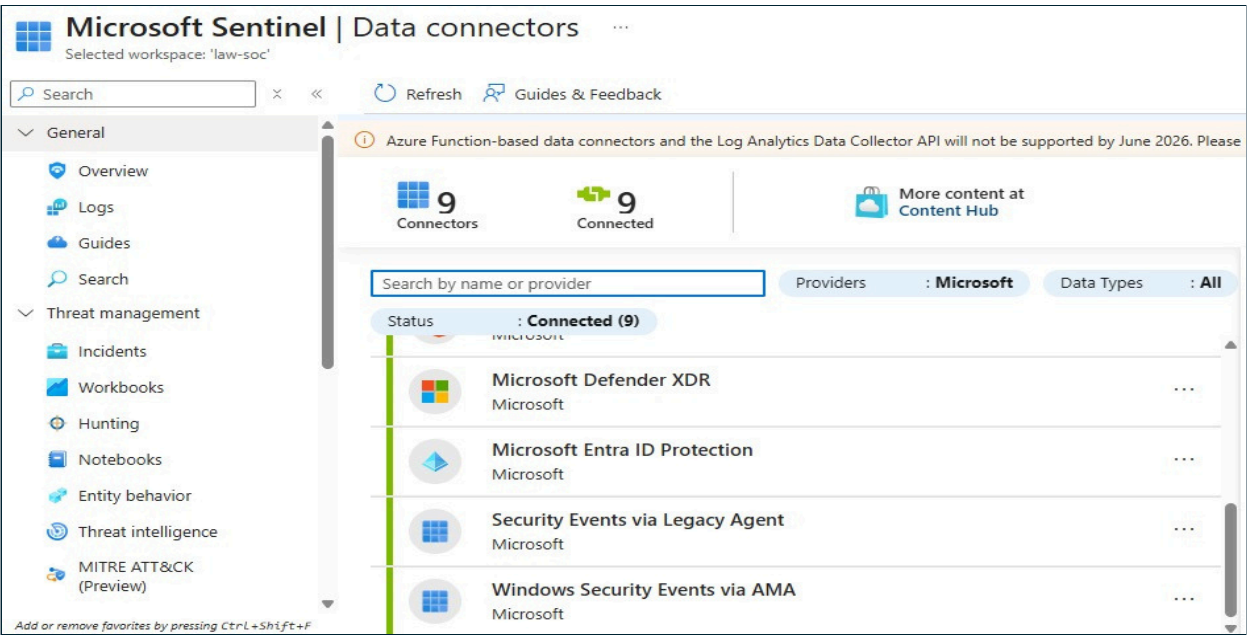


Figure D.9 — SecurityEvent Logs

Screenshot of collected security logs.

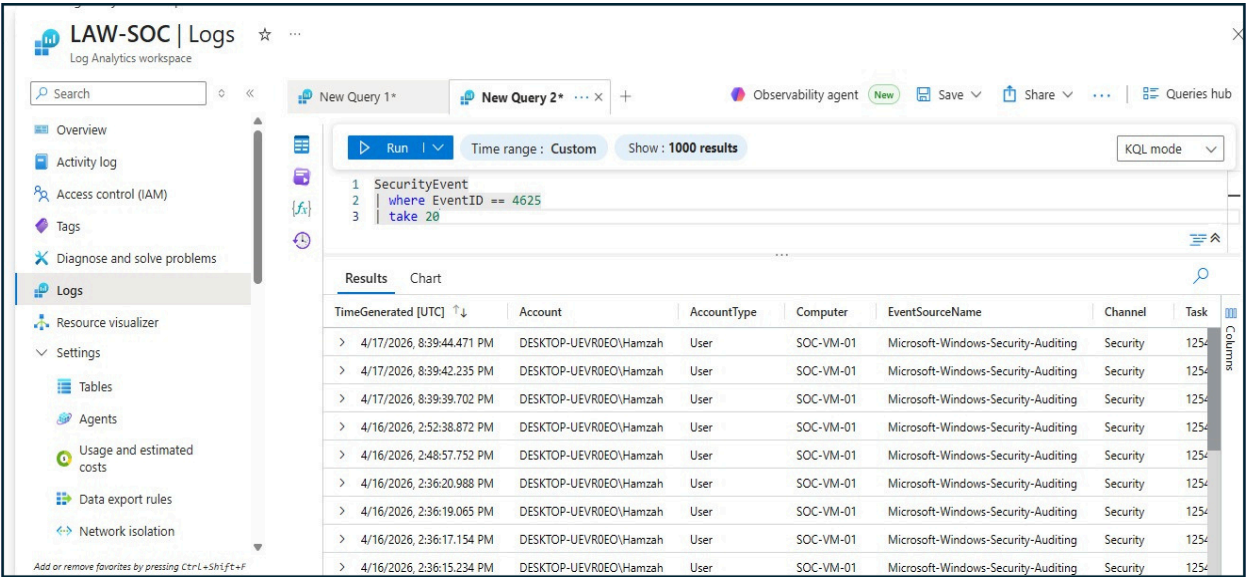


Figure D.10 — Workbook Dashboard

Screenshot of monitoring workbook.

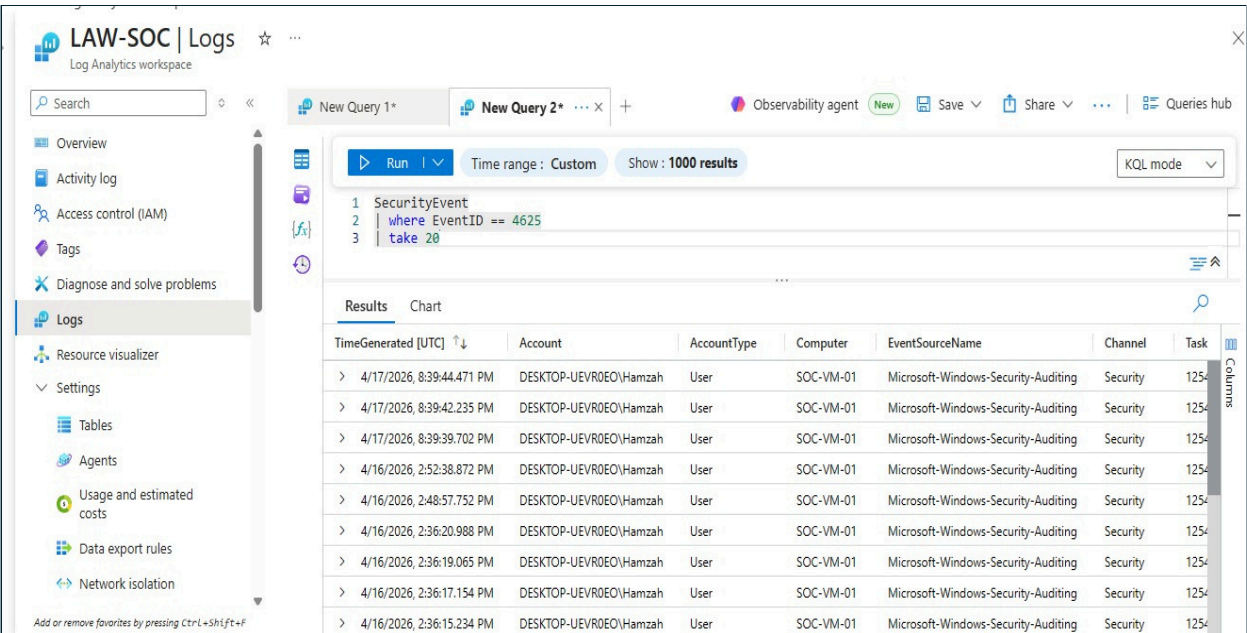


Figure D.11 — Failed Login Events

Screenshot showing Event ID 4625.

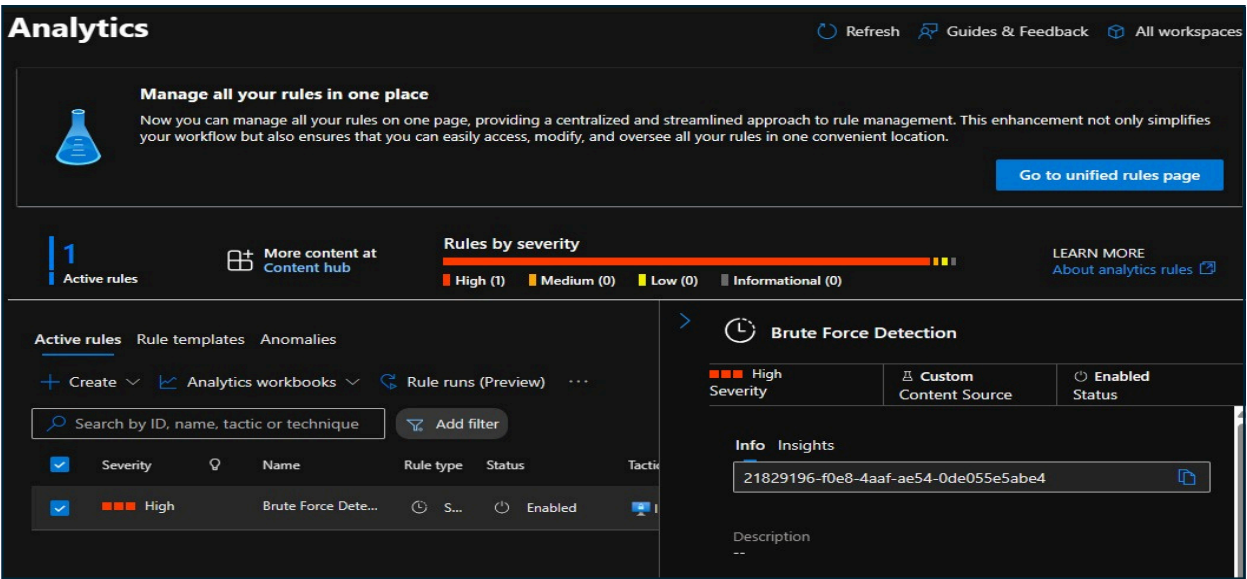


Figure D.12 — Analytics Rule

Screenshot of detection rule.

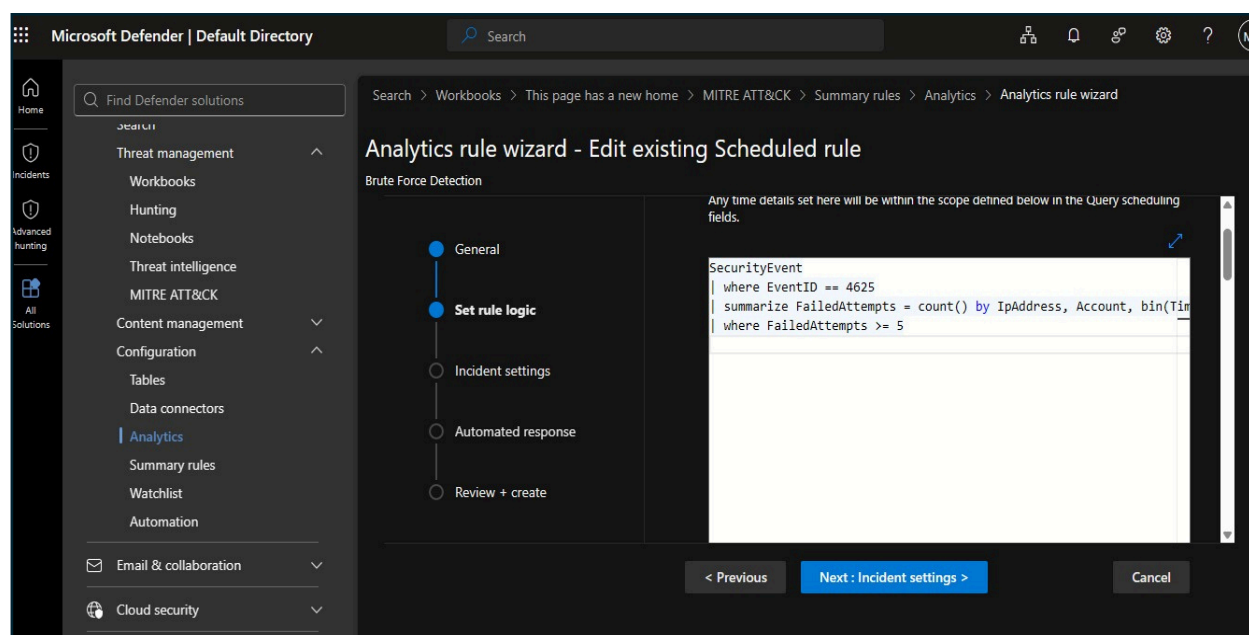
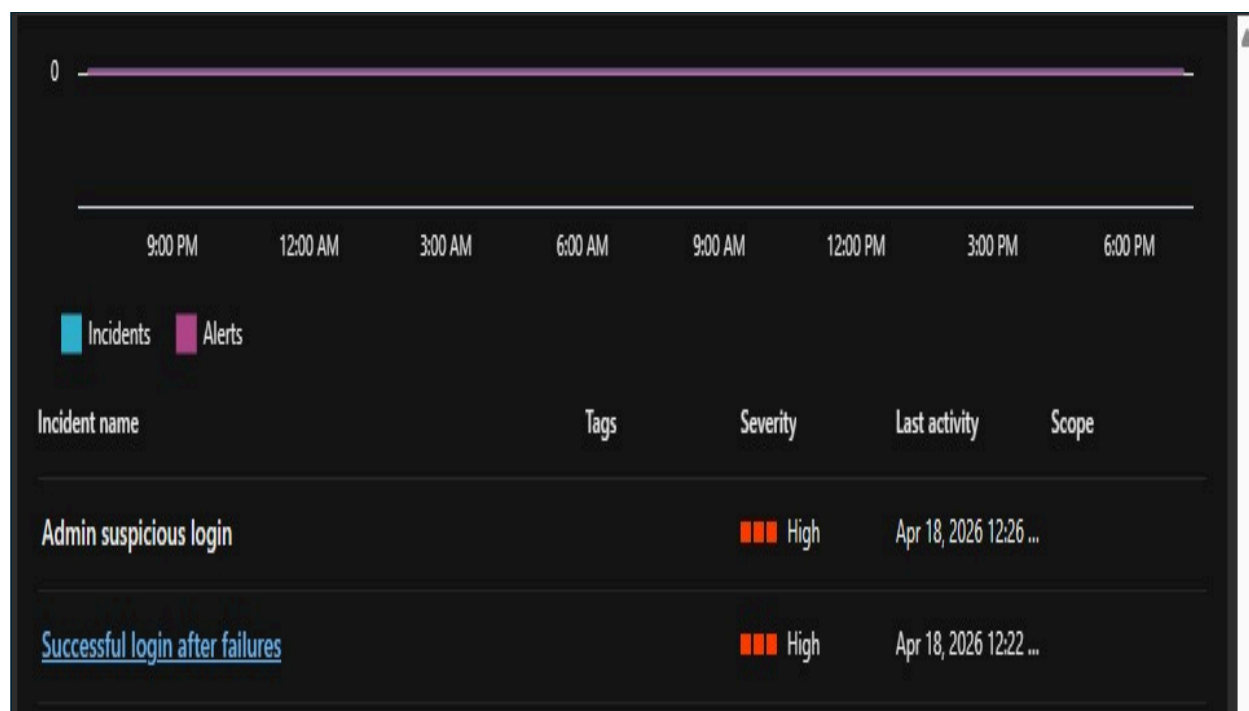


Figure D.13 — Incident Creation

Screenshot showing generated incidents.



## REFERENCES

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- [4] Scarfone, K., Mell, P. Guide to Intrusion Detection and Prevention Systems. NIST Publication.
- [5] Fyodor. Nmap Network Scanning Guide. Insecure.Org.